



Application of conditional generative adversarial network in ground motion modelling encompassing epistemic uncertainty

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Abstract Most existing Ground Motion Models (GMMs) are developed by specifying empirical functional forms derived from expert judgment, with regression techniques used to estimate model coefficients from recorded ground motion data. However, due to the complex and nonlinear nature of earthquake source and path effects, it is difficult to establish a direct analytical relationship between ground motion intensity measures and seismological predictors. This, combined with inherent natural variability, poses significant challenges for accurately characterizing ground motion using predefined functional forms. The emergence of deep learning methods offers a promising avenue for uncovering non-linear correlations among high-dimensional variables. In this study, the deep learning technique, Conditional Generative Adversarial Network (CGAN), is proposed as a novel, data-driven GMM for predicting horizontal-component spectral accelerations over a period range of 0 to 10 s. The model is trained and evaluated using 11675 sets of recorded ground motions from the Engineering Strong motion database (ESM2.0). Model performance is rigorously evaluated through multiple metrics, including residual analysis and comparison with benchmark empirical

GMMs. Results demonstrate that the CGAN outperforms traditional models in capturing complex spectral patterns and exhibits superior generalization with reduced prediction bias. Furthermore, a comparative analysis with other deterministic and probabilistic machine learning models developed using the same dataset highlights similarity in aleatory uncertainty but notable differences in epistemic uncertainty estimation, attributed to the fundamentally different uncertainty quantification mechanisms. A key strength of the CGAN approach lies in its ability to generate physically consistent, high-fidelity synthetic ground motion spectra, making it a promising alternative to standard regression-based GMMs and conventional deep learning models.

Keywords Conditional generative adversarial network · Engineering strong motion database · Ground motion models · Mixed effect regression · Epistemic uncertainty

1 Introduction

Seismic design largely depends on the response spectrum representation of ground motion, most important being peak ground acceleration (PGA) and elastic response spectral acceleration for various structural periods ranging from 0.1 ~ 10 s. Given that the predictor (Magnitude, source-to-site distance, site class, etc.) are identified for the characterization

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of target variables (Spectral ordinates), a predefined functional form based on expert judgement is traditionally used to represent mathematical relationships and the physics between the predictor and target variables. Regressions are carried out using the spectral ordinate of recorded ground motions and the seismological parameters to determine the coefficients of the regression in the functional form. The assumed function may not accurately capture the entirety of the actual situation. Such imperfect functional forms may, thus, introduce model deficiencies leading to prediction errors. Alternatively, a non-parametric model based on machine learning algorithms can provide a better alternative as it does not require a predefined functional form. Artificial neural networks (ANNs) known for their effectiveness in learning and generalizing data relationships, have recently found widespread application in seismic engineering and seismology. Typically, ANNs consist of neural networks with one or more hidden layers between the input and output layers. ANNs may be inadequate or constrained by numerous limitations when it comes to capturing correlations among multiple variables (Ding et al. 2021).

Further, traditional feedforward neural networks as well as models like Random Forests, Genetic Algorithms etc. are useful for point predictions and basic uncertainty estimates. However, they cannot capture the full conditional distribution of spectral amplitudes, offering only limited probabilistic insight (e.g., mean and aleatory standard deviation) rather than the complete variability in seismic ground motion. The emergence of deep learning methods, however, presents a robust solution. Deep learning (or Deep neural networks, DNN), advanced forms of ANN, proves highly adept at simulating the distribution of high-dimensional data and recognizing intricate nonlinear relationships. Furthermore, the data-driven approaches enable deep learning models to effectively comprehend complex seismic nonlinearities inherent in the data without necessitating separate feature extraction steps. Widely used variants of DNNs include Bayesian Neural Networks (BNNs), Variational Autoencoders (VAEs), and Generative Adversarial Networks (GANs), each applied across diverse domains. For instance, Fayaz et al. (2025) proposed a hybrid framework for earthquake early warning (EEW) that integrates a domain-informed VAE, Gaussian Process Regression, and explainable AI to

enable real-time spectral ordinate prediction. Likewise, Fayaz and Galasso (2022) developed a VAE-based on-site spectral ordinate estimator leveraging early P-wave signals and site conditions for real-time applications. Yedulla et al. (2025) developed a BNN-based ground motion model (GMM) for horizontal and vertical-to-horizontal (V/H) spectral components using the Engineering Strong Motion Database (ESM 2.0) and quantified the associated epistemic uncertainty. Using the same dataset, Sriwastav et al. (2025) develop a GMM using Conditional VAE.

BNNs incorporate epistemic uncertainty by placing prior distributions (e.g., Gaussian priors) over the network weights to approximate the posterior distribution. The epistemic uncertainty in BNNs is quantified through the variance of the predictive posterior distribution. While BNNs effectively capture parameter uncertainty, their ability to model complex, multimodal conditional distributions is limited. CVAEs model the conditional distribution $P(y|x)$ by learning a latent variable model. The encoder maps the input x and output y to a latent space z , while the decoder reconstructs y from x and z . The objective function for CVAEs combines a reconstruction loss with a Kullback–Leibler (KL) divergence term to regularize the latent space. CVAEs capture epistemic uncertainty through the stochasticity of the latent variables, but their unimodal Gaussian assumptions in the latent space can restrict their ability to represent complex, nonlinear relationships in ground motion data. CGANs, by contrast, model the conditional distribution $P(y|x)$ without explicit assumptions about its functional form. The adversarial training process enables CGANs to learn highly nonlinear and multimodal distributions, making them particularly suited for capturing the complex variability in ground motion IMs.

This study primarily explores a variant of GAN for the prediction of spectral ordinates for horizontal components of ground motion. GAN is based on the principle that all physical phenomena can be explained through noise. When source noise undergoes transformation via a medium, represented by a transfer function, it ultimately manifests into the observable output. In an analogous sense with earthquakes, one may assume that an earthquake generates a noise that gets filtered as it traverses through layered media, resulting in a signal of specific bandwidth. Thus, GAN can be viewed as a model characterized by randomness, distinguishing it from other deterministic DNNs. GAN

consists of two simultaneous networks, Generator and Discriminator, capable of learning and replicating the underlying probability distribution of the training data, enabling the generation of synthetic samples that are statistically similar to real observations. Lei et al. (2020) illustrated that a GAN can learn the intrinsic manifold of analyzed data from a geometric standpoint. When augmented with auxiliary input variables, GAN are extended to Conditional GAN or CGAN, wherein both the generator and discriminator are conditioned on additional information to guide the data generation process (Mirza and Osindero 2014). Michelsanti and Tan (2017) utilized a CGAN model to enhance the performance of a speech system in noisy environments, surpassing classical methods in terms of efficacy. Xiong and Chen (2019) employed CGAN to generate load time histories for walking, jumping, or bouncing actions based on class labels, effectively modeling human loads.

In seismic aspects, GAN and its variants have been used in earthquake early warning systems, seismic data (waveform) augmentation, Geophysical studies such as Interpolation, Super sampling, Ground-roll, De-noising, Data compression, Seismic inversion, and Distributed Acoustic Sensing (DAS) systems. An extensive review of the nearly 150 published literatures between 2018–2022 on the application of GAN and its variants in earthquake engineering and geophysical studies is presented by Marano et al. (2024), indicating its popularity and widespread application. However, there are comparatively limited works in ground motion modeling using GANs. Ding et al. (2021) used DNN and CGANs to predict aftershock spectral accelerations utilizing inputs mainshock Spectral ordinates at 21 periods and eight seismic variables including M_w , R_{JB} of main shock, Hypocenter depth (km), V_{s30} and fault mechanism. It was observed that the CGAN model slightly outperformed the DNN model due to its ability to generate new data randomly. Wang et al. (2021) applied GAN to Oklahoma seismic data, demonstrating that synthetic waveform augmentation enhances earthquake detection, particularly with limited labeled training data. Esfahani et al. (2023) utilized CGAN to construct a predictive model capable of simulating nonstationary ground motion time histories in the time–frequency domain. Shi et al. (2024) introduced a conditional ground motion synthesis method based on a Generative Adversarial Neural Operator, which operates

independently of the sampling frequency of the input ground motion records. Shen et al. (2024) proposed two models to directly generate aftershock time histories from mainshock time histories, employing a convolutional network as the foundational network layer and a CGAN as the overarching structure. The outcomes demonstrated the efficacy of the proposed models in generating aftershock acceleration time histories closely aligned with the observed trends. Comparative analysis of intensity measures and response spectra revealed that the two models surpass the mainstream model (Abrahamson et al. 2014, ASK14).

Neelamraju et al. (2024) developed a CGAN-based model for predicting response spectra using the NGA-West2 database. Huang et al. (2024) presented a CGAN-based framework for simulating ground motions across varying magnitudes, distances, and site conditions using a hybrid dataset combining recorded and stochastically simulated accelerograms for training. Ba et al. (2025) introduced a spectrum-compatible ground motion simulation model using CGAN. The model incorporates a response spectrum discriminator and key conditional inputs to ensure both spectral compatibility and physical realism in generated acceleration time histories. Lin and Hu (2025) presented a CGAN model designed to generate near-fault ground motions conditioned on engineering parameters. Ding et al. (2025) introduced a deep generative model based on CGAN for simulating three-component ground motion intensity envelope functions. Dalmasso et al. (2025) utilized CGAN to create a hybrid dataset that replicates realistic seismic responses and enables binary classification of structural behavior.

In GMM development, standard ANN or DNN models lack inherent uncertainty quantification. In contrast, CGANs generate output ensembles for given inputs, capturing both epistemic and aleatory uncertainties. This generative nature enables comprehensive assessment of total variability in ground motion predictions. Despite the recent studies employing CGANs in ground motion modeling, a comprehensive, non-parametric GMM tailored specifically for the seismically active European region and surrounding areas remains unavailable. CGAN is employed in this paper to develop a non-parametric GMM for spectral accelerations (S_a) of horizontal components for a period range of 0 to 10 s. The training and evaluation data encompassed 11,675 sets of records from

the Engineering Strong Motion (ESM2.0) database. To capture the complex, hierarchical variability in ground motions, a cross-nested mixed-effects regression structure is integrated, modeling inter-event, inter-site, inter-locality, and inter-region random effects. The capability of CGAN framework in the generation of multiple outputs per input and explicit estimation of both epistemic and aleatory uncertainties are also demonstrated in this paper. Following section presents details of the ground motion dataset used for the development of GMM in this paper.

2 Selection of ground motion dataset

In this study, a subset of the ESM2.0 database used by Sriwastav et al. (2024) serves as the foundation, and the same record exclusion criteria from that study are applied here. Events with a focal depth ≤ 40 km are considered. Site characteristics are represented by V_{s30} , average shear wave velocity of the top 30 m soil layer. The Joyner-Boore distance (R_{JB}), served as the distance measure. Additionally, events that lacked either M_w , R_{JB} , V_{s30} , or focal depth (d) are also removed. A total of 11,675 ground motions falling into the following range were selected: M_w 4.0–7.8, $R_{JB} < 400$ km, focal depth (d) < 40 km and V_{s30}

between 180–1500 m/s. These records account for 1,464 events and 1,402 stations across Europe and its surrounding regions. Figure 1 presents the spatial distribution of seismic events where as Fig. 2 depicts the $M_w - R_{JB}$ distribution and frequency distribution of the ground motion records with M_w , R_{JB} , V_{s30} , d , and fault category. The data exhibits a non-uniform and skewed distribution, potentially complicating linear regression analysis. Notably, there are not enough near-field ground motion recordings, as well as data between M_w 7.0–7.5, which necessitates the employment of non-linear regression techniques.

Further, Fig. 3 presents the distribution of seismic events and sites in the dataset. Figure 3a shows a strong concentration of unique seismic events in lower magnitude bins, with 265 events in the 5.0–5.5 M_w range and a steep decline at higher magnitudes with only 4 events each in the 7.0–7.5 and 7.5–8.0 bins, indicating a significant imbalance that may limit modeling accuracy for large earthquakes. Figure 3b highlights a similar imbalance in site conditions, with the majority of recording sites falling within the 360–560 m/s V_{s30} range, while $V_{s30} > 1300$ m/s are represented by only 6 locations, suggesting potential bias in site response modeling. The 2D histogram in Fig. 3c further emphasizes that the dataset is densely populated in the mid-magnitude (M_w 4.5–6.5) and

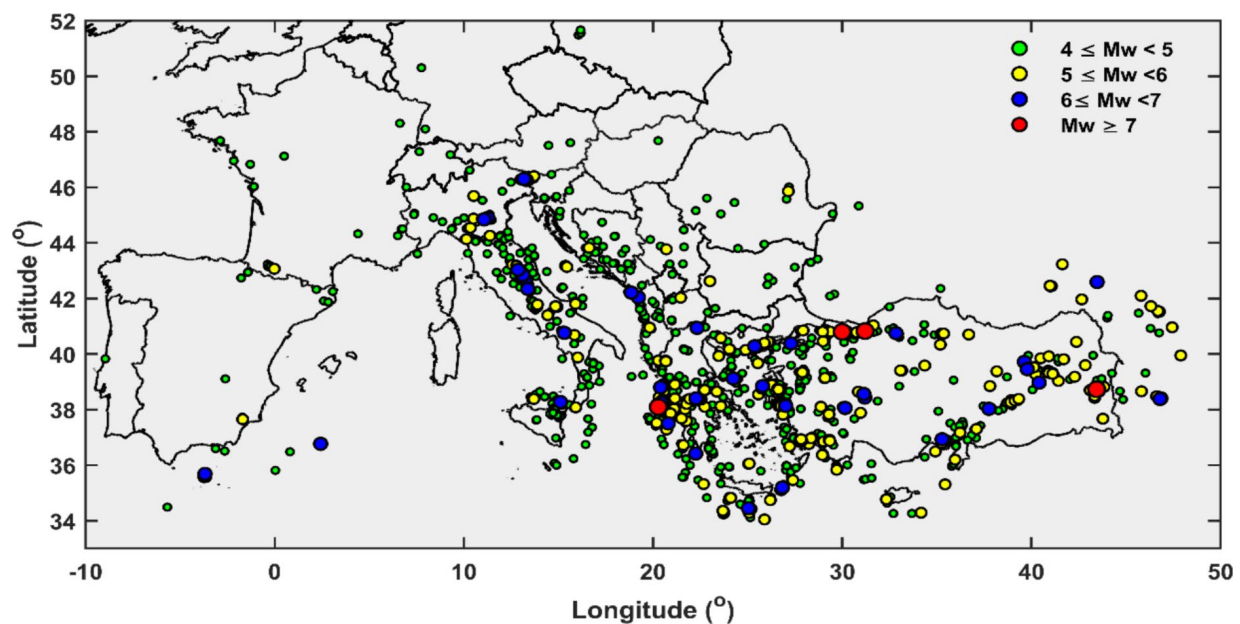


Fig. 1 Spatial distribution of the epicenter of the considered seismic events

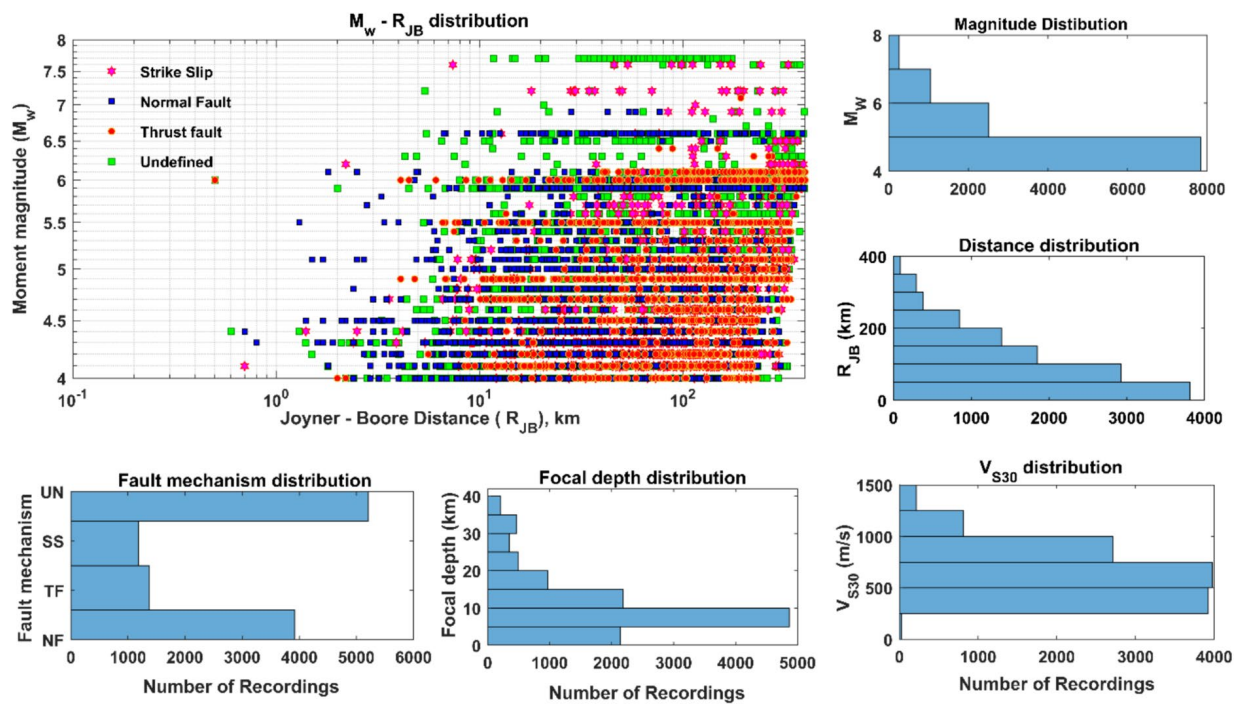


Fig. 2 $M_w - R_{JB}$ distribution and histogram of records with seismological parameters M_w , R_{JB} , V_{S30} , d , and fault category

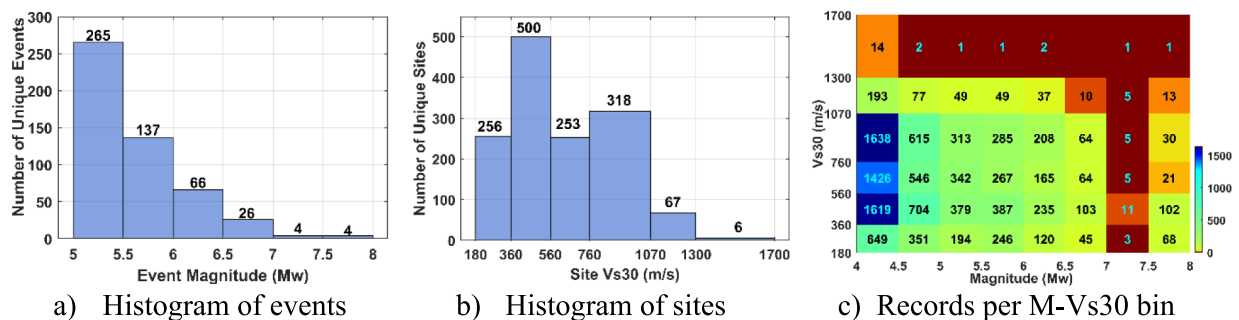


Fig. 3 Histogram of events, Histogram of sites and number of records associated with event site bin

mid- V_{S30} (360–760 m/s) regions, while combinations involving large-magnitude events and hard rock conditions are underrepresented, with some bins having fewer than five records.

Prior to analysis, all the ground motions undergo band pass filtering with 8th-order Butterworth filter with a frequency range of 0.1 to 20 Hz. This band can capture the frequency range most relevant to structural response, as periods between 0.05 s and 10 s encompass typical engineering applications. Frequencies above 20 Hz have limited influence on structural performance and are often dominated by noise. The

resulting acceleration time series are then demeaned, and cosine tapering is applied to the initial and final 5% of the signal. The following section provides a discussion on CGAN.

3 Conditional generative adversarial network (CGAN)

GANs are a class of artificial intelligence models introduced by Goodfellow et al. (2014). GANs consist of two neural networks: a generator and a

discriminator. The generator (denoted as G hereafter) produces synthetic data samples from random noise using multiple layers of neural units, whereas the discriminator (denoted as D hereafter) is a binary classifier that distinguishes between real and generated samples, aiming to classify real samples as 1 and samples generated by G as 0. Over multiple epochs, the generator develops the ability to generate samples that are indistinguishable from real ones. G serves as a mapping function that transforms input noise variables $z \sim P_z$ (typically white noise drawn from a uniform or normal distribution) into synthetic data $G(z; \theta_g)$ that approximates samples drawn from the real data distribution P_{data} . Here, θ_g denotes the learnable parameters (weights and biases) of the generator network. Simultaneously, the discriminator $D(x; \theta_d)$ functions as a binary classifier that attempts to distinguish between real samples $x \sim P_{data}$ and fake samples $x' = G(z) \sim P_{model}$, with θ_d representing

its learnable parameters. The GAN training process is formulated as a two-player minimax optimization problem to reach a Nash equilibrium, where the generator strives to accurately model the data sample distribution to generate realistic data samples with the value function $V(D, G)$ given by:

$$\max_G \min_D V(D, G) = E_{x \sim P_{data}} [\log D(x)] + E_{z \sim P_z} [\log(1 - D(G(z)))] \quad (1)$$

At the Nash equilibrium, the generator G has learned a distribution $P_{model} \approx P_{data}$, making the discriminator unable to distinguish real from generated data better than random guessing, i.e., $D(x) \approx 0.5$. With additional information (y) such as M_w, R_{JB}, V_{S30} etc., introduced to both G and D , GAN extends to a CGAN. The inputs to the generator are now pairs (z, y) , and the discriminator receives pairs (x, y) , aiming to distinguish real vs. generated samples conditioned on y . The CGAN objective becomes:

$$\max_G \min_D V(D, G) = E_{(x,y) \sim P_{data}} [\log D(x, y)] + E_{(z,y) \sim P_z, P_y} [\log(1 - D(G(z, y), y))] \quad (2)$$

In this case, $G(z, y; \theta_g)$ generates conditioned Sa samples and $D(x, y; \theta_d)$ outputs the probability that x is real, given the condition y . Figure 4 illustrates the typical structure of a CGAN. $z \in \mathbb{R}^d$ is a latent vector representing the inputs of G , comprising independent random variables following a uniform distribution in the range of -1 to 1 . $P_z \sim U(-1, 1)^d$ denotes the white noise distribution. $y \in \mathbb{R}$ represents seismological parameters serving as the conditions for the CGAN. These parameters to be learned during optimization include the weights and biases of the neural network layers for the generator and discriminator, respectively.

The loss function in the Eq. (2) serves as the core mechanism guiding the training of both the generator and the discriminator. The discriminator loss maximizes the likelihood of correctly identifying real

versus synthetic data, while the generator loss minimizes the discriminator's ability to do so. The training procedure of the CGAN begins by sampling a minibatch of m noise vectors $\{z^{(i)}\}_{i=1}^m \sim P_z$. Simultaneously, a corresponding minibatch of conditioning vectors $\{y^{(i)}\}_{i=1}^m \sim P_y$, representing physical parameters such as magnitude or site conditions, is drawn. These inputs are then passed to the generator G , producing a set of synthetic outputs $\hat{x}^{(i)} = G(z^{(i)}, y^{(i)}; \theta_g)$. To guide adversarial learning, a minibatch of real samples $\{x^{(i)}\} \sim P_{data}$ is also sampled. The discriminator parameters θ_d are updated via gradient descent using the discriminator loss $\nabla_{\theta_d} L_D$, which seeks to improve the model's ability to distinguish real from generated samples. Given the parameters of the discriminator $\theta_d = [w_{d1}, w_{d2}, b_{d1} \dots]$

$$\nabla_{\theta_d} L_D = \left[\frac{\delta L_D}{\delta w_{d1}}, \frac{\delta L_D}{\delta w_{d2}}, \frac{\delta L_D}{\delta b_{d1}}, \dots \right]; L_D = -E_{(x,y) \sim P_{data}} [\log D(x, y)] - E_{(z,y) \sim P_z, P_y} [\log(1 - D(G(z, y), y))]; \quad (3)$$

Following this, the generator parameters θ_g are updated based on the generator loss $\nabla_{\theta_g} L_G$, encouraging the generator to produce more realistic data

conditioned on y . Assuming the generator has parameters $\theta_g = [w_{g1}, w_{g2}, b_{g1} \dots]$, then,

$$\nabla_{\theta_g} L_G = \left[\frac{\delta L_G}{\delta w_{g1}}, \frac{\delta L_G}{\delta w_{g2}}, \frac{\delta L_G}{\delta b_{g1}}, \dots \right]; L_G = -E_{(z,y) \sim P_z P_y} [\log(1 - D(G(z, y), y))]; \quad (4)$$

This iterative training continues until both networks converge, ideally reaching a Nash equilibrium. Model development, optimization and prediction of horizontal spectral ordinate using the CGAN framework is provided in the following section (Table 1).

4 Prediction of spectral ordinates using CGAN

The model employs six predictor variables representing key seismic and geotechnical parameters, selected for their relevance to ground motion

variability. These inputs were identified through a rigorous feature selection process involving sensitivity analysis and statistical relevance (e.g., mutual information, permutation importance). The selected features effectively capture critical contributors to ground motion behavior, including source characteristics, path effects, and site-specific conditions. These parameters include the M_w , R_{JB} , $\log R_{JB}$, $\log V_{s30}$, focal depth, and the style of faulting. The target variables are the logarithmic transformations of PGA and Sa at 36 discrete periods, which span from 0.01 to 10 s. Logarithmic transformation is applied as these parameters follow a lognormal distribution. This transformation normalizes the data, reducing skewness and making it more suitable for machine learning models, which perform better with normally distributed data. The specific periods corresponding to each Sa value are detailed in Table 2. The formulation, including the predictor and descriptor variables, is provided in Eq. (5).

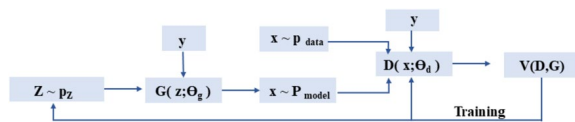


Fig. 4 Block diagram of a conditional generative adversarial network (CGAN)

Table 1 Sample hyperparameter tuning results

Configuration	A	B	C	D	E	F
Layers (G/D)	3, 3	3, 3	4, 4	3, 3	3, 3	2, 2
Neurons G	30–40–50	30–40–50	30–40–50–60	30–40–50	30–40–50	30–40
Neurons D	50–40–30	50–40–30	60–50–40–30	60–50–40	50–40–30	40–30
Learning rate	1.0E-04	1.0E-03	1.0E-04	1.0E-04	5.0E-05	1.0E-04
Activation	LeakyReLU	LeakyReLU	Tanh	LeakyReLU	LeakyReLU	ReLU
Dropout	0.3	0.2	0.4	0.3	0.3	0.3
Batch size	256	256	128	64	64	32
Batch Normalization	Minmax	Standard scaler	Minmax	Standard scaler	Standard scaler	Minmax
Epochs	3000	2000	100,000	2000	2000	50,000
MSE	0.167	0.179	0.24	0.16	0.171	0.21
R ²	0.859	0.834	0.815	0.871	0.846	0.821
Comments	Good trade-off, converged well	Low error, slightly over-fitting	Mode collapse observed, large epoch required	Best trade-off, converged well	Good trade-off, converged well	Stable, moderate performance, large epoch required
Physics Satisfied?	Yes	Yes	No	Yes	Yes	Yes

Table 2 Performance indices

* N is the number of data points; y_i is the observed (true) value at instance i ; $\hat{y}_i$ is the predicted value at instance i ; $\bar{y}_i$ is the mean of the observed values; $\bar{\hat{y}}_i$ Mean of the predicted values; σ_y is the standard deviation of the observed values;

Index	Equation
Correlation coefficient (R)	$R = \frac{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}}_i)(y_i - \bar{y}_i)}{\sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}}_i)^2 \sum_{i=1}^N (y_i - \bar{y}_i)^2}}$
Root mean-squared error ($RMSE$)	$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$
Mean absolute error (MAE)	$MAE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)$
Mean absolute percentage error ($MAPE$)	$MAPE = \frac{1}{N} \sum_{i=1}^N \left(\left \frac{y_i - \hat{y}_i}{y_i} \right \right)$
Performance Parameter (PP)	$PP = 1 - \frac{RMSE^2}{\sigma_y}$

$$\log[Sa_{T_0 \sim T_{10}}] \sim f(M_w, R_{JB}, \log(R_{JB}), \log(V_{S30}), focal\ depth, fault\ flag) + error \quad (5)$$

$Sa_{T_0 \sim T_{10}}$ denotes the spectral ordinates associated with different periods. For fault mechanism, flag 1 is given for Normal faulting (NF), 2 for Thrusting (TF), 3 for Strike-slip (SS), and 4 for Unknown.

4.1 Model development

The modeling and training of CGAN is conducted using TensorFlow, a deep learning framework developed by Abadi et al. (2016). TensorFlow offers extensive Python-based utilities for building, training, and

optimizing deep learning models, facilitating efficient model development, experimentation, and deployment across diverse applications. Figure 5a depicts the typical structure of GAN with white noise (latent space) as input for G , the recorded Sa at different periods serves as an input target along with generated Sa sample for D , which discriminates between recorded and generated data by assigning values of 0 and 1 to the generated and recorded data, respectively. Figure 5b denotes the typical structure of CGAN. The primary difference between GAN and CGAN is the

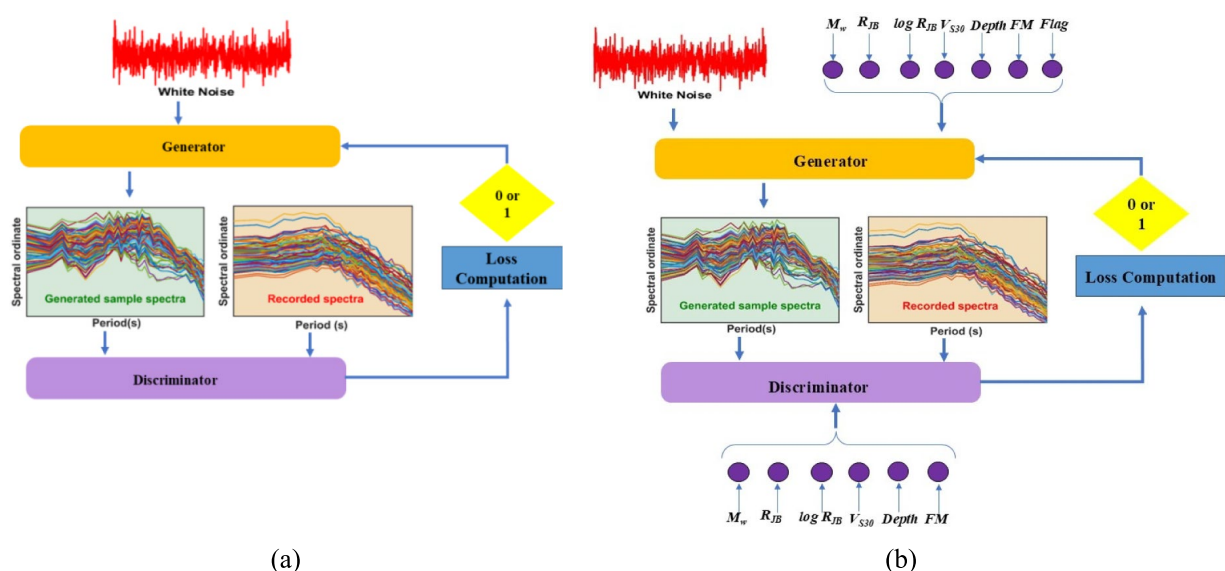


Fig. 5 a) Typical GAN architecture generating spectra from noise. b) Typical CGAN architecture generating spectra from noise and conditional inputs

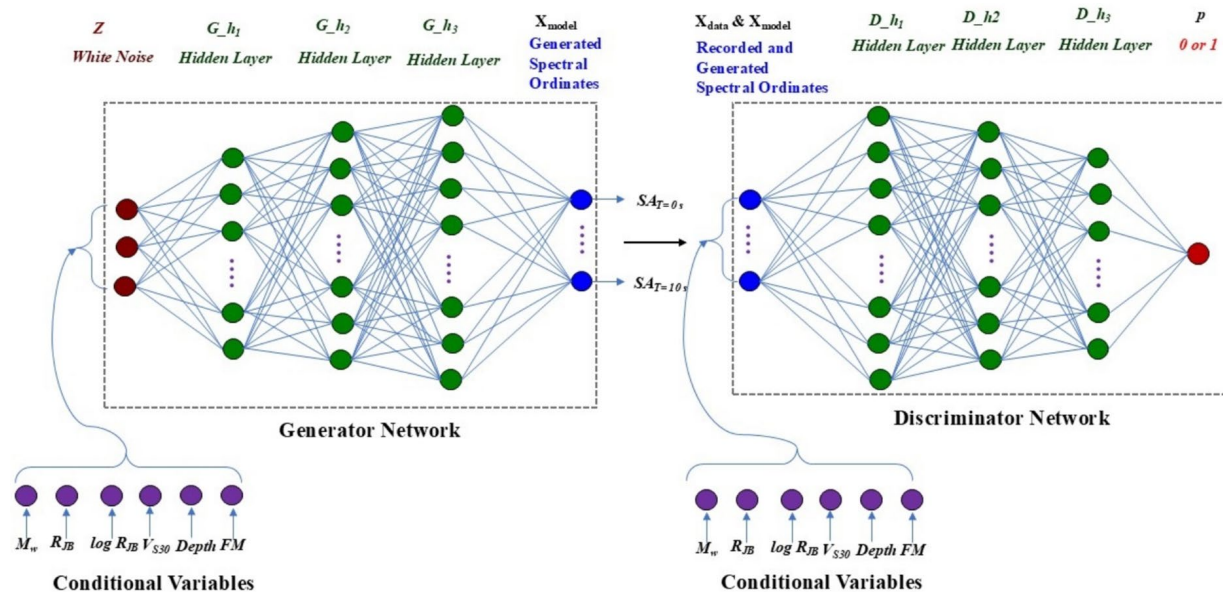


Fig. 6 Optimized architecture of the conditional generative adversarial network (CGAN) for the considered ESM dataset

additional predictor variable information provided to condition the results. In this study, the information of M_w , R_{JB} , $\log R_{JB}$, $\log V_{s30}$ Focal depth and Focal mechanism are provided as conditional inputs to the G and D .

The CGAN architecture, illustrated in Fig. 6, consists G and D networks designed and fine-tuned for robust performance in predicting spectral ordinates. G (left panel of Fig. 6) accepts a latent space input vector comprising three dimensions of random noise $z \sim U(-1,1)$, concatenated with conditioning variables: M_w , R_{JB} , $\log R_{JB}$, $\log V_{s30}$, and focal depth and the focal mechanism. Higher dimensions of z led to instability, mode collapse, and overparameterization without improving spectral diversity or realism, making three a balanced and efficient choice. These conditional inputs are incorporated only in the first layer of the Generator to modulate the noise distribution and enhance data realism from the outset.

4.2 Model optimization

Weight initialization in the trial model followed Glorot and Bengio (2010) initialization to maintain variance across layers. The loss functions for G and D follow the standard CGAN formulation, with gradients computed using backpropagation via the chain rule across all trainable weights and biases. The dataset

is systematically divided into three mutually exclusive subsets: 70% for training, 15% for validation, and 15% for testing. The model incorporated batch normalization to stabilize learning by normalizing layer inputs within each mini-batch. Preprocessing trials with *MinMaxScaler* and *StandardScaler* showed better performance with the latter; hence, input features were standardized to zero mean and unit variance.

To efficiently optimize the CGAN without an excessive number of trial runs, a structured, hierarchical hyperparameter tuning approach was employed. Parameters were tuned sequentially based on their impact on performance. Rather than the generator and discriminator losses, mean squared error (MSE) and R^2 are utilized for model evaluation. Further, Wasserstein loss, though effective for general distribution alignment, is avoided due to its added complexity (e.g., critic replacement, strict Lipschitz constraints) and its limited suitability for modeling conditional, Gaussian-like output structures. The process began by optimizing the number of layers and neurons in both generator and discriminator, followed by tuning the learning rate and activation functions. Once the core architecture was fixed, regularization parameters like dropout rate and batch normalization were refined. Training-related parameters such as batch size and number of epochs were then adjusted. For each trial, configurations are evaluated while monitoring issues

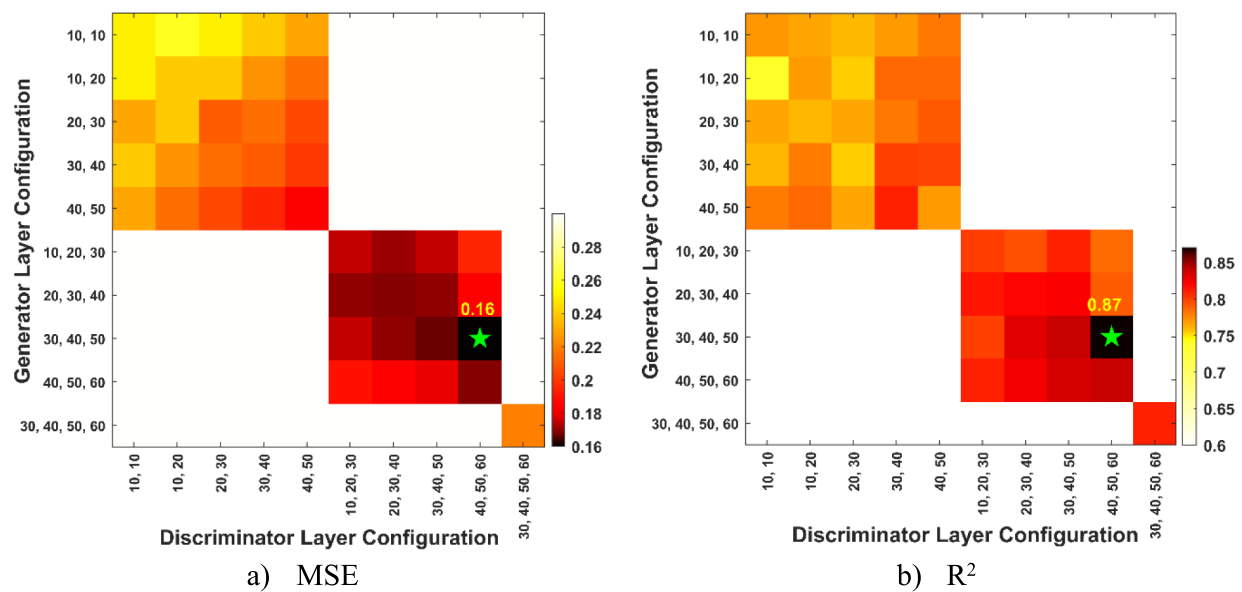


Fig. 7 Heatmap of MSE and R^2 for different CGAN generator and discriminator layer configurations

like mode collapse and vanishing gradients as well as whether the desired physics is satisfied. Sample results are presented in Table 1.

This heatmap in Fig. 7 shows the MSE and R^2 of various CGAN architectures based on G and D configurations. The architecture of model best performance (MSE=0.16, R^2 =0.87) is marked by a green star. The *G* (left side of Fig. 7) comprises three fully connected hidden layers (G_{h_1} , G_{h_2} , G_{h_3}) with 30, 40, and 50 neurons, respectively. This progressive increase in the number of neurons facilitates hierarchical feature representation, where the earlier layers model low-level abstractions and deeper layers capture complex, higher-order dependencies between the latent space and conditioning seismological variables. The decision to use three layers is based on a trade-off between underfitting (too shallow) and overfitting or vanishing gradients (too deep). The final output layer of *G* maps to the target dimension of the ground motion parameter being generated. The *D* (right side of Fig. 7) mirrors a similar conditional structure with the same seismological parameters appended to the input layer, ensuring the network evaluates generated or real samples in context. *D* is structured with three hidden layers (D_{h_1} , D_{h_2} , D_{h_3}) containing 60, 50, and 40 neurons. The output layer is a single neuron yielding a probability score for binary classification—real (1) or generated (0).

Leaky ReLU with a negative slope coefficient ($\alpha=0.2$) was adopted across all network layers. Unlike other activation functions such as sigmoid, tanh, and standard ReLU—which required epochs as high as 100,000 and more than 24 h per trial—Leaky ReLU achieved comparable or improved results in just 2,000 epochs, reducing training time to 30–60 min per trial, while leading to better model performance (Fig. 8). Another reason for choosing Leaky ReLU over standard ReLU is its ability to prevent the ‘dying ReLU’ problem, where neurons become inactive and stop updating during backpropagation. By allowing a small, non-zero gradient for negative inputs, it ensured all neurons remain active. The relatively high slope value ($\alpha=0.2$), larger than the typical 0.01, enhanced gradient flow, especially in early training when negative activations were more frequent. This is supported by Maas et al. (2013), which demonstrated that larger α values in Leaky ReLU improved learning dynamics and convergence speed. Moreover, ablation studies performed during hyperparameter tuning in this work showed that $\alpha=0.2$ yielded faster convergence (30–40% fewer epochs) and better stability compared to smaller α values, thereby justifying its use in the final architecture.

Figure 9 illustrates the impact of key hyperparameters, learning rate, batch size, and dropout rate, on the CGAN model’s performance. An

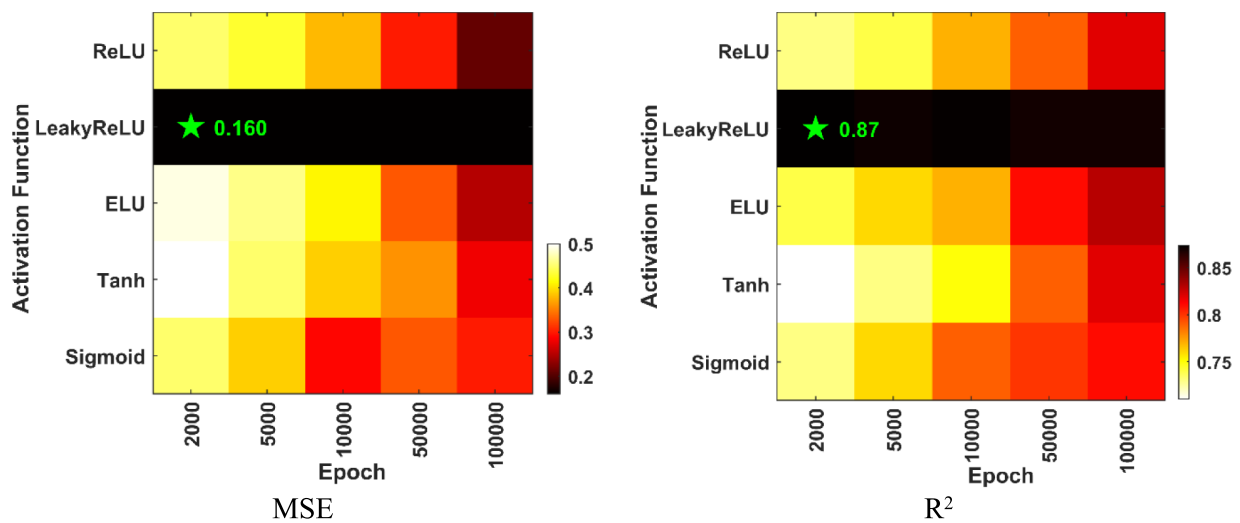


Fig. 8 Hyperparameter tuning of Epoch and activation function

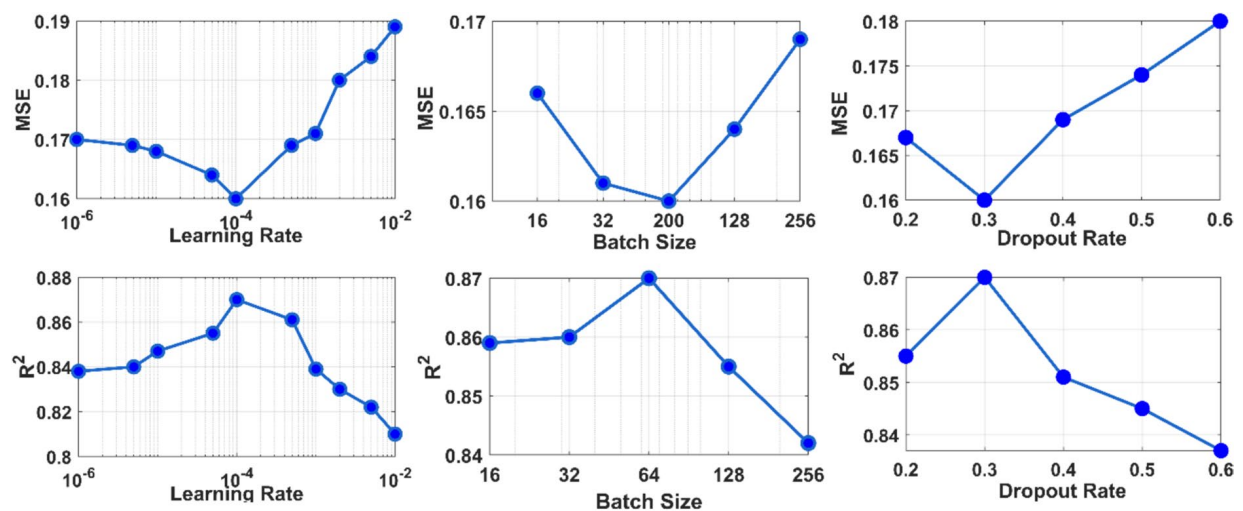


Fig. 9 Hyperparameter tuning of learning rate, batch size and dropout rate

optimal learning rate around 10^{-4} resulted in the lowest MSE and highest R^2 . The learning rates for both the generator and discriminator were set to 10^{-4} , with the Adam optimizer configured using $\beta_1 = 0.5$ and $\beta_2 = 0.999$, consistent with best practices in GAN training. For batch size, a value of 64 yields the best results. Regarding dropout rate, lower values (around 0.2–0.3) perform better, while higher rates degrade performance, likely due

to excessive regularization. To mitigate overfitting, dropout with a rate of 0.3 was applied, along with L2 regularization (weight decay = 10^{-4}) across applicable layers. This optimized architecture and hyperparameter configuration provided stable convergence and high-quality synthetic data generation that closely mimicked the statistical properties of spectral ordinate associated with real (recorded) seismic records. The optimized model's predictive

capability and statistical consistency are assessed based on the results presented in the subsequent section.

5 Mixed-effect regression

GMMs developed using machine learning frameworks often utilize mixed-effects regression to improve both accuracy and generalization. Mixed-effects models integrate fixed effects to capture global trends via explanatory variables and random effects to account for unexplained variability, reflecting differences among seismic events, recording sites, or regional geological conditions. Traditionally, mixed-effects formulations group residuals by events and stations to isolate variability specific to those clusters, allowing for unbiased and reliable predictions. In this study, the model extends this concept by introducing additional random groupings based on tectonic zones—representing distinct seismic characteristics of geologic regions, and attenuation regions—which capture spatial variability in wave energy dissipation. Incorporating both standard and extended random effects into the CGAN-based GMM enhances regression accuracy by isolating global trends from localized variability, thereby reducing predictive uncertainty and improving the model's capacity to deliver spatially adaptive and statistically robust estimates of spectral ordinates.

5.1 Event localization & tectonic locality

The variability between earthquakes is captured by the between-event random variance, with event-specific median residuals $[\delta B_e \sim N(0, \tau^2)]$ estimated for each event 'e' within this grouping. An additional random effect is employed here to quantify the variation between earthquake locations. Following Kotha et al. (2020, 2022), which utilized the European Seismic Hazard Model 2020 (ESHM20) to assign shallow crustal events to seismotectonic zones ('TECTO'), 56 distinct tectonic localities are delineated with an additional zone flag for data not falling within any of these zones. The random effects group for locality 'l' is denoted as "between-locality" $[\delta L2L_l \sim N(0, \tau_{l2l}^2)]$. This grouping is introduced to quantify earthquake locality-to-locality variability in ground motions within the dataset.

5.2 Site regionalization & attenuation regions

Site-to-site median residuals $[\delta S2S_s \sim N(0, \varphi_{s2s}^2)]$, associated with site 's', have traditionally been used to account for random effects related to site-specific characteristics. However, recent GMMs have highlighted significant differences in anelastic attenuation across regions, partly due to variations in crustal properties. Kotha et al. (2020, 2022) employed a geological-geophysical regionalization method developed by Basili et al. (2019) as part of the TSUMAPS-NEAM project. This group of random effects concerning regionality is labeled as "between-region" $[\delta R2R_r \sim N(0, \varphi_{r2r}^2)]$, symbolized by region *r*. The random variance of the attenuation regions group quantifies observed variability in anelastic attenuation between regions, with each region serving as a level. Depending on the granularity of regionalization, the estimate of between-region variance may vary. Here, 46 attenuation regions identified by Kotha et al. (2020) are utilized, along with an additional zone for data not falling within any of the 46 zones.

Given that 'locality' encompasses seismic events and 'regions' include recording stations, the model uses nested mixed-effect grouping. Further, recordings of an event can spread across multiple regions, a crossed-effect component is added, resulting in a crossed-nested mixed-type model (Stafford 2014).

Residuals for each of the random effect group are assumed to be normally distributed with mean equal to zero. The standard deviations for between-event, site, locality, and region effects are denoted by $\tau, \varphi_{s2s}, \tau_{L2L}$, and τ_{R2R} , respectively. The standard deviation of the unaccounted (intra) residuals is represented by φ . The term for between-event effects, independent of locality, $\delta B_{e,l}^0 \sim N(0, \tau_0^2)$, is given by $\delta B_{e,l}^0 = (\delta B_e - \delta L2L_l)$. Total sigma, assuming ergodicity, is provided in Eq. (6), and the between-event variance is represented by $\tau = \sqrt{\tau_0^2 + \tau_{L2L}^2}$. For non-ergodic calculations, τ_{R2R} can be included.

$$\sigma_s^2 = \tau_0^2 + \tau_{L2L}^2 + \varphi_{s2s}^2 + \varphi^2 \quad (6)$$

The mixed-effects regression is carried out based on Abrahamson and Youngs (1992) [updated by Mohammedi et al. (2023)], in a way similar to Derras et al. (2014), involving a repeated fitting procedure. Specifically, the neural network is first trained to predict the target parameter, after which the mixed-effects

regression is applied to the residuals to estimate random effects. These effects are then used to adjust the data, and the model is retrained. This iterative process continues until convergence is achieved. The process

is detailed below. The structure is adapted from Kotha et al. (2020, 2022), with a nonlinear model-based function $f_{model}(\cdot)$. Specifically, the natural logarithm of the observed ground motion $GM_{e,l,s,r}$ is modelled as

$$\ln(GM_{e,l,s,r}) \sim N\left(f_{model}(X_{e,l,s,r};\theta) + \left|\delta B_{e,l}^0, \delta L2L_l, \delta S2S_s, \delta R2R_r\right|, \varphi^2\right) \quad (7)$$

$f_{model}(X_{e,l,s,r};\theta)$ is model-predicted median ordinate as a nonlinear function of predictors X trained to capture the deterministic part. $\left|\delta B_{e,l}^0, \delta L2L_l, \delta S2S_s, \delta R2R_r\right|$ indicate that these residuals cannot be directly added. This hierarchical model includes nested effects and crossed effects. The model is implemented in iterative phases:

- i. *Fixed-Effects via model*: The model is trained to predict $Y_{e,l,s,r}^i \sim f_{model}(X_{e,l,s,r}^i; \theta)$, minimizing the negative log-likelihood loss under the assumption that residuals follow a Gaussian distribution.

- ii. *Random-Effects Modeling*: The residuals from the model $\delta_{e,l,s,r}^i = Y_{e,l,s,r}^i - f_{model}(X_{e,l,s,r}^i; \theta)$, are then decomposed into the structured random intercepts via a mixed-effects model:

$$\delta_{e,l,s,r}^i = \left|\delta B_{e,l}^0, \delta L2L_l, \delta S2S_s, \delta R2R_r\right| + \varepsilon_i \quad (8)$$

where ε_i is the residual error not explained by the random effects. It is implemented in MATLAB using the *fitlme* function, which allows specification of both nested and crossed random effects using formula syntax. The specific part of the MATLAB code is presented below.

`dataset = table(res, Locality, Event, Region, Site);`

`lme = fitlme(dataset, 'res ~ 1 + (1|Locality) + (1|Locality/Event) + (1|Region) + (1|Region/Site)');`

where, the ‘dataset’ includes residuals (*res*) along with categorical grouping variables: *Event*, *Locality*, *Site*, and *Region*. The ‘lme’ object contains the fitted linear mixed-effects model, including estimates of fixed and random effects, variance components, residuals, and statistical summaries that quantify variability due to event, site, locality, and region.

- iii. A new model is then trained using a fixed-effect procedure on the adjusted target $\left(Y_{e,l,s,r}^i - \delta_{e,l,s,r}^i\right)$.
- iv. Steps (i) through (iii) are iterated until convergence is achieved, as indicated by a less than 0.5% difference between successive log-likelihood values. The relative change in the log-likelihood (LH) between iterations is monitored using Eq. (9) and terminate the iterative process when log-likelihood ratio (LHR) falls below 0.5%, indicating convergence.

$$LHR = \frac{LH(ii) - LH(ii - 1)}{LH(ii - 1)} 100\% \quad (9)$$

The trained model is provided as collab files (see *Data and Resources* section). The detailed performance evaluation and regression results provided in the following section.

6 Model performance

This section presents a detailed evaluation of the model performance using recorded ground motion data. The analysis encompasses regression diagnostics, including fixed-effect coefficients, random-effect realizations, residual distributions, and associated uncertainty metrics. In subSection 6.1, the performance of the network is discussed in detail. SubSection 6.2 analyzes the behavior of fixed-effect components to understand the relationship between

predictors like magnitude and distance intervals with the target variables. SubSection 6.3 examines random effects and residuals for any biases or trends with predictor variables including the discussion on the aleatory and epistemic uncertainty of the developed model.

6.1 Network performance

The network's performance is assessed using five metrics: Correlation Coefficient (R), Root Mean-Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Performance Parameter (PP) (Table 1). The R value indicates the linear relationship between predicted and actual values. RMSE and MAE quantify the average magnitude of prediction errors, with RMSE giving more weight to larger errors. MAPE offers a perspective on error relative to actual values. PP consolidates multiple performance aspects into a single measure. Together, these metrics evaluate the accuracy of the regression model, supporting its assessment and refinement. The CGAN generates spectral predictions by mapping Gaussian noise through a learned transformation, introducing inherent stochastic variability and inter-period non-smoothness. To obtain stable and representative estimates, the model is executed 100 times per input scenario, and the median response across realizations is used for evaluating performance and computing ground motion prediction metrics. The metrics calculated between the median CGAN prediction and the target for 37 spectral ordinates, using the finalized network parameters and hyperparameters, are detailed in Table 3. Remarkably, all periods show an R value exceeding 0.89, demonstrating a strong linear correlation between predicted and recorded Sa, and confirming the model's high predictive accuracy. MAPE values remain stable across the period range, with no evident monotonic trend, indicating uniform model performance irrespective of spectral period. Furthermore, the PP value exceeds 0.8, underscoring the model's overall effectiveness in predicting the target spectral ordinates.

6.2 Fixed effects

The physics of the predicted spectral ordinates with the target values across different fixed-effect predictors is evaluated here. As noted previously, these

Table 3 Performance of developed CGAN model

T	R	RMSE	MAE	MAPE	PP
log(Sa) 0.00 s (g)	0.905	0.973	0.765	1.948	0.813
log(Sa) 0.010 s (g)	0.905	0.960	0.755	1.899	0.818
log(Sa) 0.025 s (g)	0.906	0.977	0.771	2.414	0.814
log(Sa) 0.04 s (g)	0.906	0.989	0.781	2.486	0.816
log(Sa) 0.05 s (g)	0.904	1.019	0.806	2.068	0.810
log(Sa) 0.07 s (g)	0.902	1.038	0.821	2.804	0.810
log(Sa) 0.10 s (g)	0.895	1.058	0.837	1.950	0.801
log(Sa) 0.15 s (g)	0.892	1.042	0.826	1.774	0.801
log(Sa) 0.20 s (g)	0.891	1.017	0.801	1.701	0.811
log(Sa) 0.25 s (g)	0.892	0.991	0.778	3.147	0.800
log(Sa) 0.30 s (g)	0.892	0.985	0.772	2.408	0.801
log(Sa) 0.35 s (g)	0.892	0.971	0.760	3.644	0.810
log(Sa) 0.40 s (g)	0.896	0.936	0.729	2.386	0.802
log(Sa) 0.45 s (g)	0.896	0.929	0.725	3.492	0.803
log(Sa) 0.50 s (g)	0.895	0.932	0.728	3.920	0.800
log(Sa) 0.60 s (g)	0.897	0.927	0.725	2.963	0.802
log(Sa) 0.70 s (g)	0.900	0.922	0.719	2.088	0.805
log(Sa) 0.75 s (g)	0.901	0.908	0.705	4.093	0.811
log(Sa) 0.80 s (g)	0.899	0.928	0.724	10.840	0.803
log(Sa) 0.90 s (g)	0.901	0.920	0.714	3.363	0.808
log(Sa) 1.00 s (g)	0.904	0.908	0.705	12.971	0.814
log(Sa) 1.20 s (g)	0.908	0.896	0.693	1.708	0.822
log(Sa) 1.40 s (g)	0.911	0.905	0.705	1.732	0.822
log(Sa) 1.60 s (g)	0.915	0.874	0.677	2.127	0.838
log(Sa) 1.80 s (g)	0.918	0.887	0.688	1.629	0.836
log(Sa) 2.00 s (g)	0.919	0.871	0.677	1.221	0.845
log(Sa) 2.50 s (g)	0.921	0.884	0.689	2.155	0.845
log(Sa) 3.00 s (g)	0.925	0.885	0.687	1.352	0.850
log(Sa) 3.50 s (g)	0.928	0.873	0.677	0.994	0.857
log(Sa) 4.00 s (g)	0.931	0.863	0.669	7.728	0.863
log(Sa) 4.50 s (g)	0.932	0.853	0.659	0.848	0.867
log(Sa) 5.00 s (g)	0.933	0.848	0.653	1.205	0.870
log(Sa) 6.00 s (g)	0.936	0.838	0.645	0.582	0.875
log(Sa) 7.00 s (g)	0.935	0.843	0.652	0.556	0.873
log(Sa) 8.00 s (g)	0.937	0.831	0.638	0.971	0.877
log(Sa) 9.00 s (g)	0.935	0.846	0.652	0.375	0.871
log(Sa) 10.0 s (g)	0.933	0.852	0.657	1.714	0.869

comparisons are based on the median of the 100 ensemble predictions. While the assessment is conducted across the entire range of periods, detailed comparisons are specifically presented for spectral ordinates at 0, 0.2, 1, and 10 s to maintain brevity. Figure 10 presents predicted spectral ordinates as a function of moment magnitude across varying

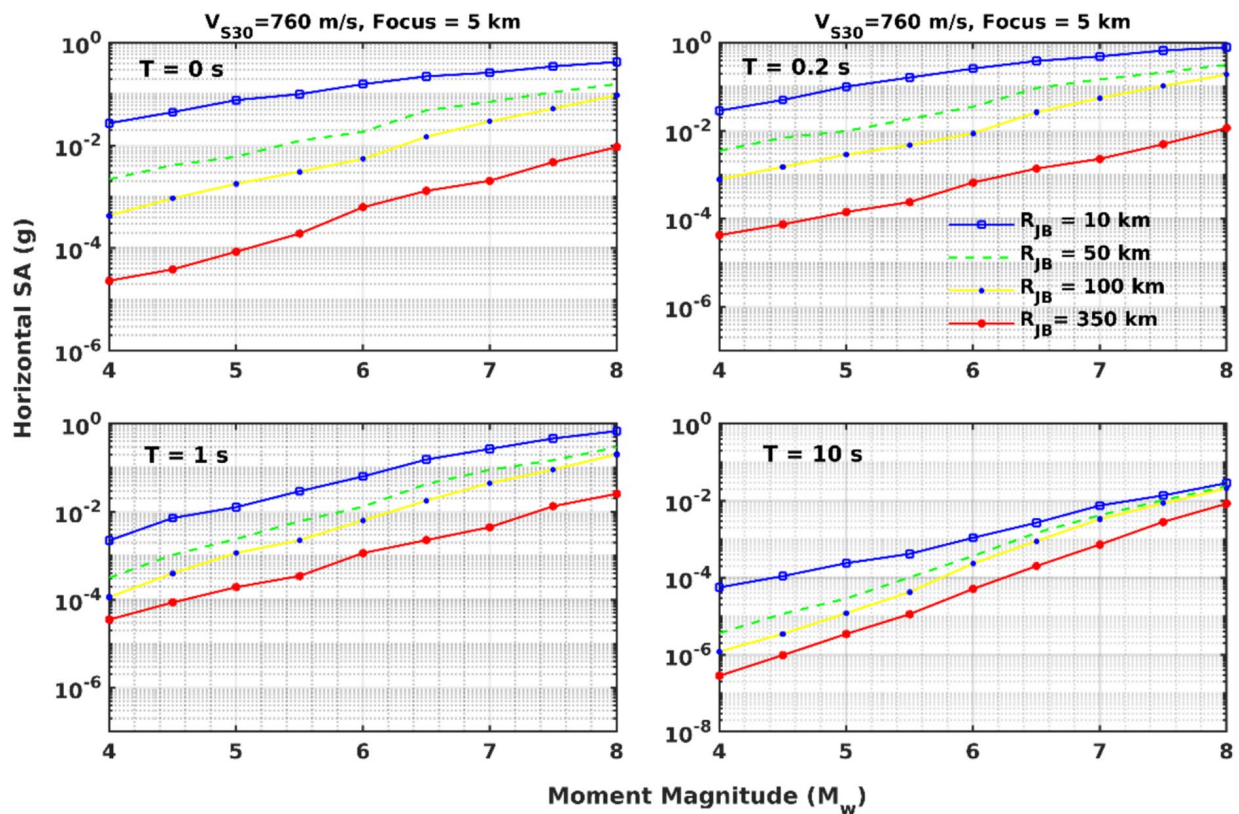


Fig. 10 Variations of spectral ordinates with M_w at different periods for different R_{JB}

distances. A clear monotonic increase in S_a with magnitude is observed, aligning with physical expectations of greater energy release in larger earthquakes. At higher magnitudes, however, the growth rate diminishes, reflecting saturation. This nonlinear scaling highlights that while stronger events produce higher ground motions, the incremental increase in S_a becomes progressively smaller, consistent with empirical trends and theoretical seismic source scaling models. It may, however, be noted that the magnitude scaling at smaller magnitudes ($M_w \leq 5.5$) shows a shallower slope than at intermediate magnitudes, particularly at longer periods where spectral amplitudes are small and more sensitive to variability. This trend arises from both physical and data-driven factors. At low magnitudes and long periods, reduced spectral contrast limits resolution in the input space, causing non-parametric models like CGAN-based networks to smooth these regions. Sparse training data further restricts the model's ability to capture sharp transitions. In contrast, scaling becomes steeper

and more consistent when the data distribution is more uniform. Stability in poorly sampled ranges can improve via data augmentation or hybrid models embedding physical constraints.

Figure 11 illustrates the attenuation of spectral ordinates with increasing distance for all magnitudes. This trend reflects the physical attenuation of seismic waves, where S_a values decrease as the waves propagate away from the earthquake source due to geometric spreading and anelastic damping. The consistent reduction in S_a across different magnitudes confirms this expected seismological behavior. The model's ability to reproduce the monotonic increase in S_a with magnitude, the saturation effect at higher magnitudes, and the attenuation with distance suggests that it accurately reflects the complex relationships between seismic event characteristics and resulting ground motion.

The comparison of the predicted response spectra for different soil categories is presented in Fig. 12, showcasing various combinations of magnitude (M_w

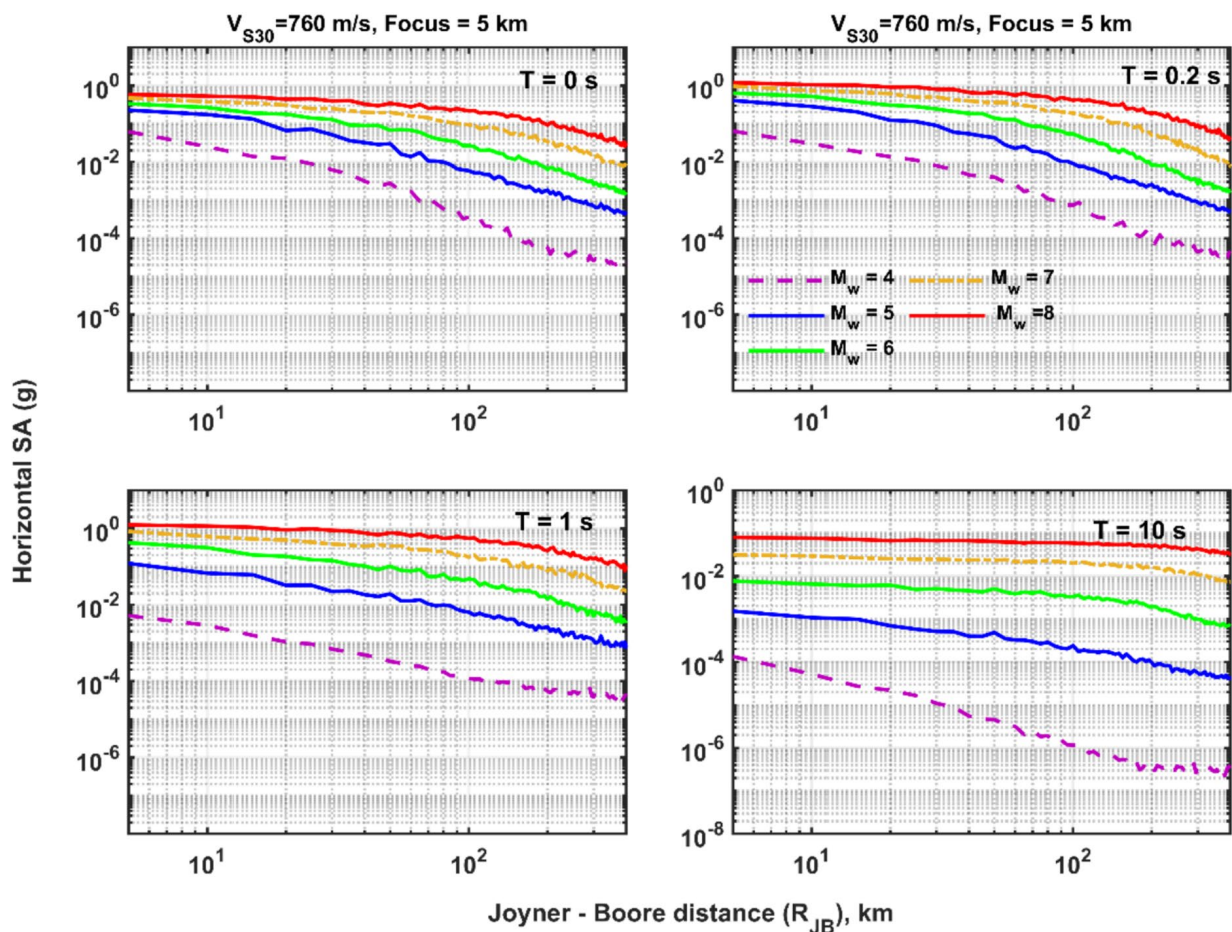
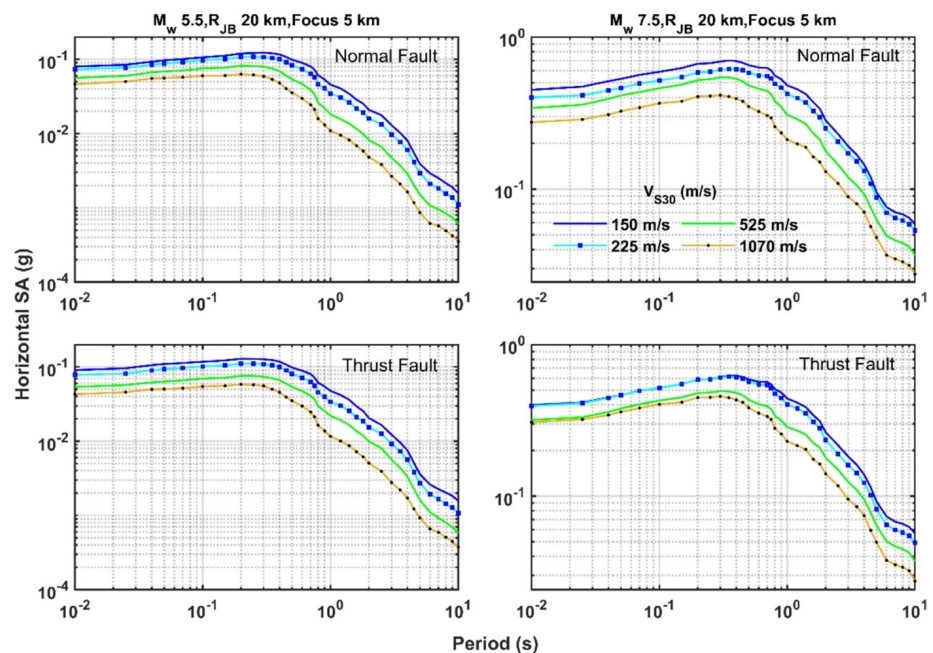


Fig. 11 Variation of spectral ordinates with R_{JB} at different periods for different M_w

5.5 and M_w 7.5) and fault mechanisms (normal and thrust faulting). The response spectra are illustrated for five different V_{S30} values: 150 m/s, 225 m/s, 525 m/s, and 1070 m/s. The observed increase in spectral amplitude with decreasing V_{S30} confirms the model's ability to capture soil amplification effects reliably. This amplification is more pronounced at higher periods because softer soils, typically associated with low V_{S30} values resonate with long-period seismic waves. An analysis of the median response-spectra ratios of M_w 7.5/ M_w 6.5 for three V_{S30} values (400 m/s, 600 m/s and 800 m/s) is presented in Figure 24 (in the Appendix). Four independent trials are performed; each trial generated 100 ensemble predictions per scenario. Owing to the CGAN's noise-induced and data-driven nature, these trials yielded visibly different median realizations. From a physical standpoint, nonlinear site response should

add damping at higher input levels, reducing amplitudes (amplification) for larger magnitudes and producing shallower magnitude scaling (i.e., lower M_w 7.5/ M_w 6.5 spectral ratios). The ASK14 model nearly exhibits this soft-site weakening trend. However, it is not apparent in Figure 24. Lacking a physics term for nonlinear site effects in the model, an adversarial objective emphasizes distributional fidelity rather than cross-derivative behavior. The inconsistencies in Figure 24 arise from data imbalance and covariate shift, with sparse large- M_w /short- R_{JB} coverage. Adversarial and stochastic training further smooths low-range regimes, inflating soft-site ratios at smaller magnitudes and longer periods. A clearer trend emerges when results are generated with 400 ensemble predictions for each M_w - V_{S30} pair (Figure 25 in the Appendix). Across four trials, the outcomes remain consistent with the 100-ensemble case

Fig. 12 Variations of response spectra with V_{S30} , M_w 5.5 (left) and M_w 7.5 (right), R_{JB} as 20 km and focal depth 5 km



but show reduced randomness and sharper patterns. The M_w 7.5/ M_w 6.5 spectral ratio (SR) decreases with lower V_{S30} , i.e., $SR(400 \text{ m/s, blue}) < SR(600 \text{ m/s, yellow}) < SR(800 \text{ m/s, green})$. With 1000 ensembles (Figure 26), this trend becomes more evident. While the model captures soil amplification, 100 ensembles are insufficient; reliable stability requires ~ 400 or more. Thus, for the proposed model, at least 400 ensemble predictions per scenario is recommended to robustly capture soil amplification. Further improvement in the CGAN model can be enforced through: (i) a physics-based penalty that encourages $\partial \ln Sa / \partial M$ to decrease for softer V_{S30} beyond an intensity threshold; (ii) hybridization with an explicit nonlinear site term (e.g., conditional on PGA or strain); (iii) re-weighting/augmenting under-represented training cells; and (iv) quantile (median) losses to stabilize conditional medians. These extensions lie beyond the scope of the present study and are not pursued further.

The variation of the response spectrum with focal depth is analyzed in Fig. 13. The results show that higher spectral amplitudes are observed for deep focus events compared to shallow ones across all periods. Figure 14 presents the variation of spectral ordinates with different focal mechanisms, demonstrating that the model captures the impact of the source mechanism on the mean spectral ordinate. Although the differences between normal and thrust

faulting mechanisms are relatively small, the model reliably reflects these subtle variations, indicating its robustness in accounting for different seismic source characteristics. Overall, these comparisons affirm that the model is reliable for estimating mean spectral ordinates based on a variety of seismological parameters. The model's ability to accurately reflect soil amplification effects, depth-dependent variations, and the influence of fault mechanisms ensures its applicability in seismic hazard assessment.

6.3 Random effects

6.3.1 Aleatory uncertainty

Accurate seismic hazard characterization requires understanding the natural randomness in ground motion. This study estimates such uncertainty using residual standard deviations, incorporating random effects via cross-nested mixed-effects regression detailed in Section 5. The variability of different residuals, such as inter-event, inter-site, and intra (remaining) concerning predictor variables, is depicted in Fig. 15. To streamline presentation, the residuals are exclusively plotted for $T=0 \text{ s}$ for brevity. Each figure incorporates error bars denoting the mean and one standard deviation bound. While inter-event and inter-site residuals generally fall

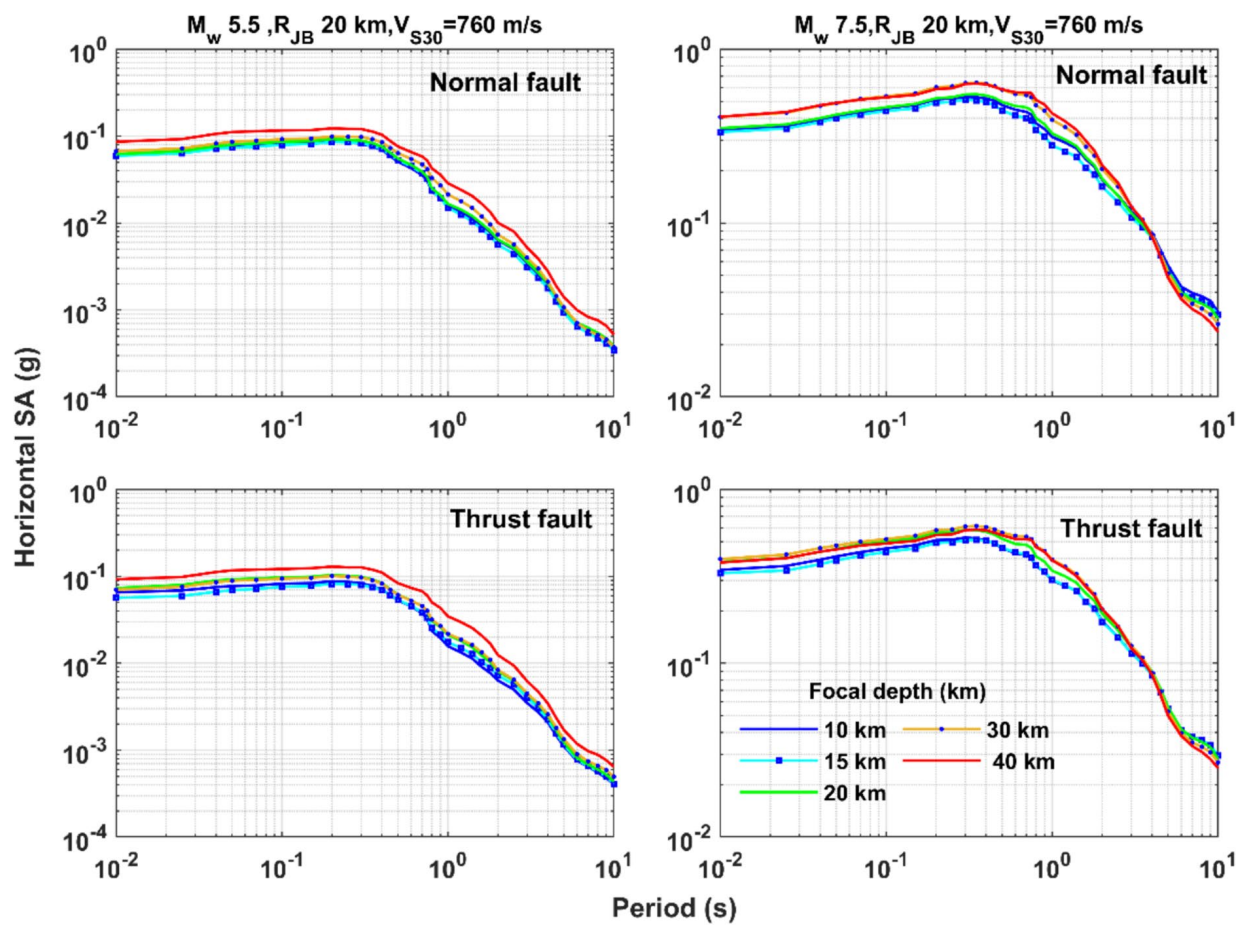


Fig. 13 Variations of response spectra with focal depth for M_w 5.5 (left) and M_w 7.5 (right), R_{JB} as 20 km and $V_{S30}=760$ m/s

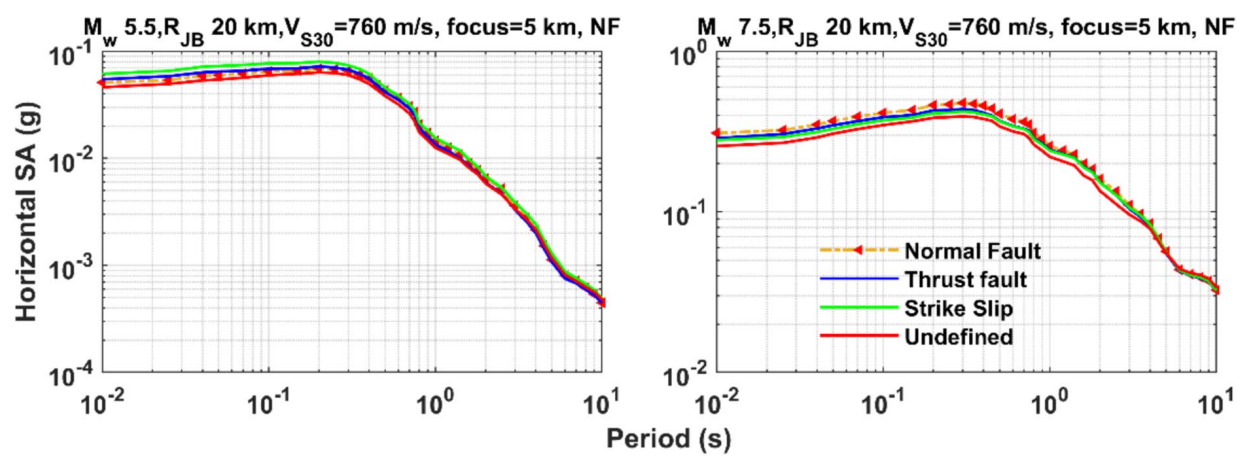


Fig. 14 Variations of response spectra with fault mechanism for M_w 5.5 (left) and M_w 7.5 (right)

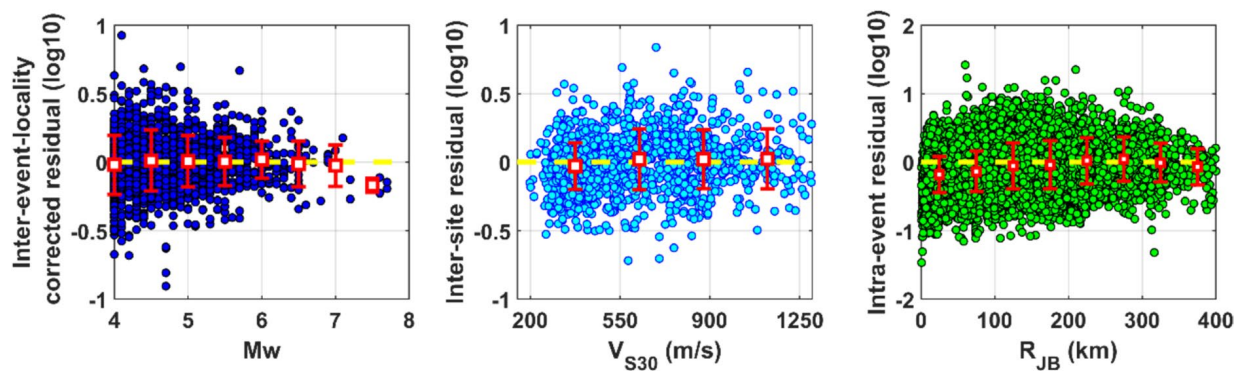


Fig. 15 Distribution of locality-corrected inter-event (left), inter-site(middle) and Intra (right) residuals for spectral ordinates at $T=0$ s

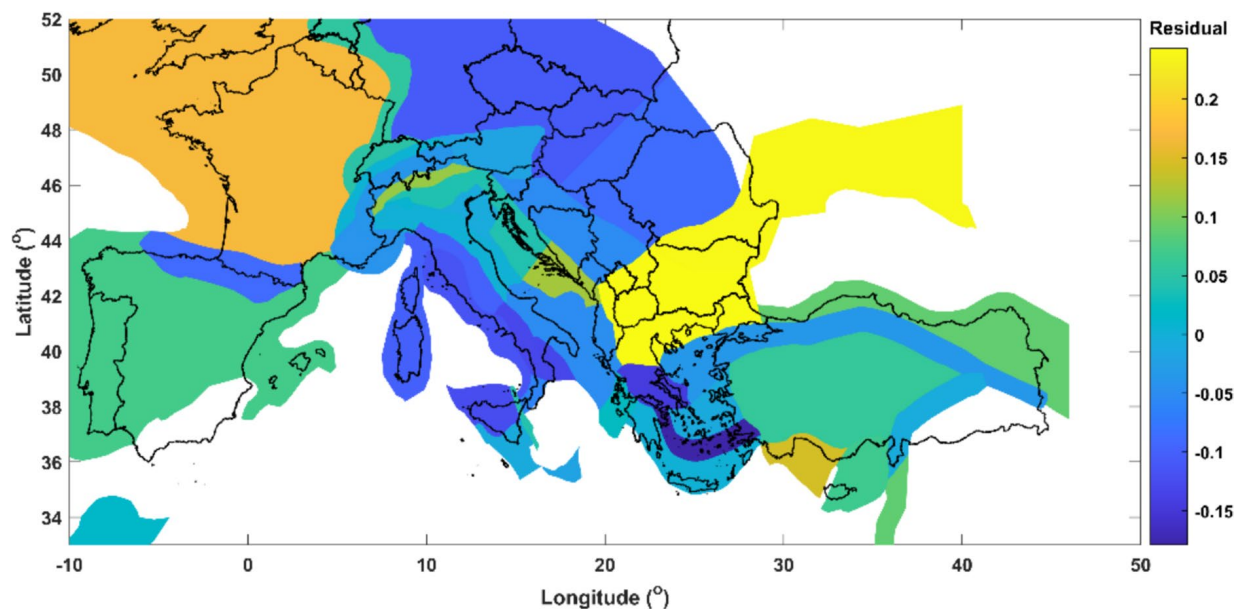


Fig. 16 Inter-region residuals across 47 attenuation regions for Sa at $T=0.5$ s (log10)

within a range of ± 1 , intra-event residuals marginally surpass this threshold for select records. Moreover, the residual plots manifest no discernible bias concerning predictor variables, with mean residuals closely centered around zero across all bins. Additionally, the residuals exhibit a normal distribution, indicative of the robustness of the CGAN model.

Figure 16 illustrates the inter-region residuals for Sa at a period of 0.5 s across 47 distinct attenuation regions. The inter-region residuals range from -0.14 to 0.25 for $T=0.5$ s. This range reflects the significant differences in prediction across various

geographical regions. This variability can be attributed to regional differences in geological and seismological conditions that affect the propagation and attenuation of seismic waves. Figure 17 presents inter-locality residuals across various tectonic localities, ranging from -0.4 to 0.15 . This broader spread, relative to inter-region residuals, highlights greater variability in ground motion across distinct tectonic settings. Such differences likely stem from localized geological conditions, faulting styles, and other seismotectonic factors not fully represented in regional-scale models.

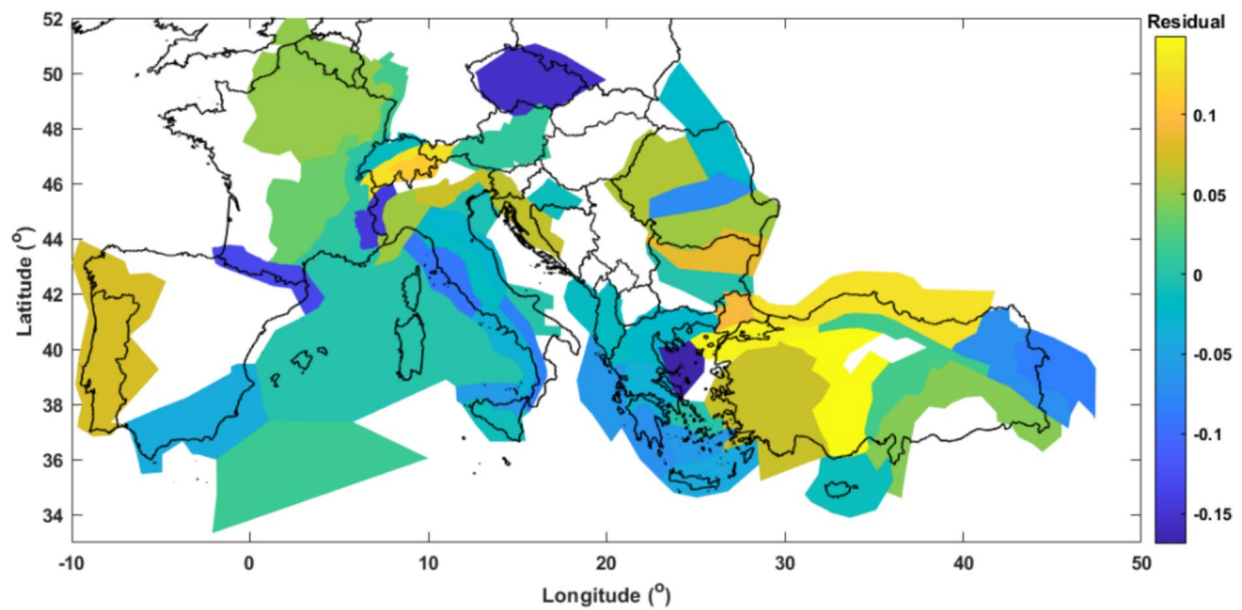


Fig. 17 Inter-locality residuals across 57 tectonic localities for SA at $T=0.5$ s (log10)

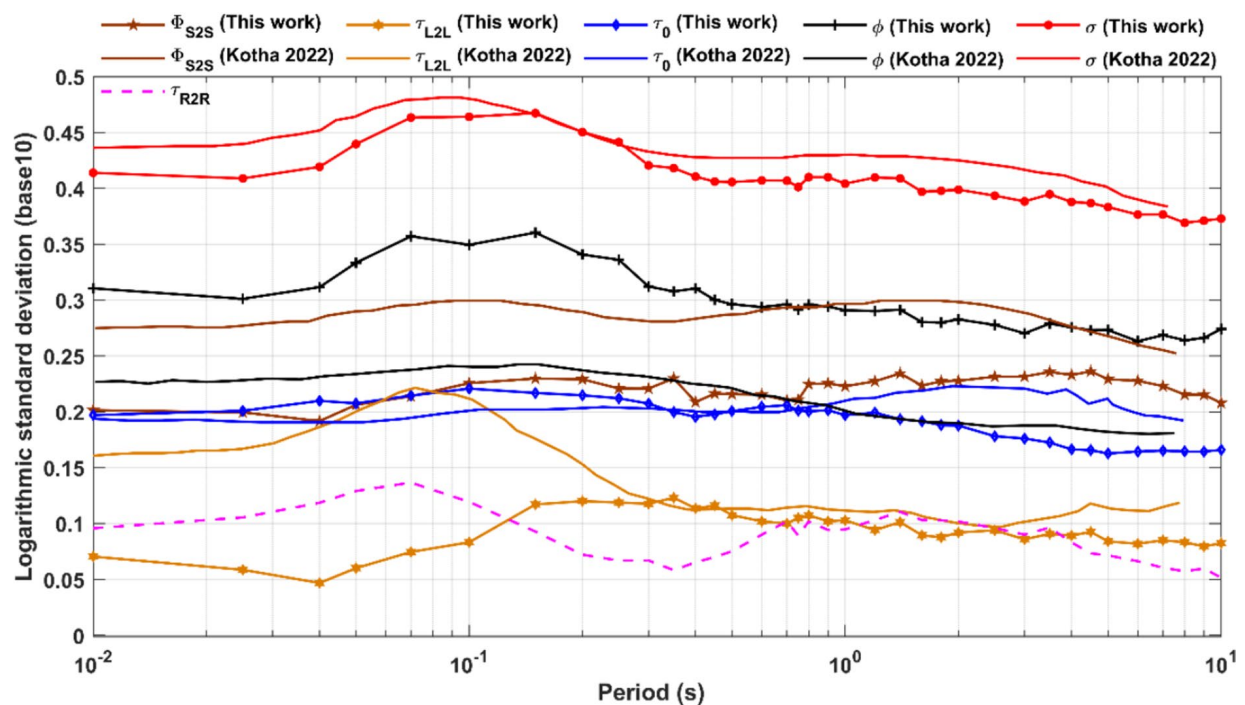


Fig. 18 Variation of different standard deviations for a horizontal component with predictor variables

Figure 18 (dashed lines) illustrates the standard deviation of all random effects (locality corrected inter-event, intra-site, inter-region, inter-locality,

and intra) for periods ranging from 0.0 to 10 s. Also presented is the total (logarithmic) ergodic standard deviation calculated using Eq. (4). Total

Ergodic sigma varies in the range of 0.36–0.47 for the horizontal component. To demonstrate the difference in the prediction of aleatory uncertainty, the standard deviations reported in Kotha et al. (2022) (referred to as Kotha2022) for the horizontal component are also included in Fig. 18. The comparison between the CGAN-based model and the regression-based model of Kotha revealed both similarities and notable differences across components of aleatory uncertainty. The inter-locality standard deviation remains nearly identical across both models, except at periods below 0.2 s, where Kotha2022 exhibits higher variability. Likewise, the locality-corrected inter-event standard deviation shows comparable trends at all periods. However, divergence is observed in the inter-site and intra components. Specifically, the inter-site standard deviation is higher in the Kotha2022, while the intra variability is more pronounced in the CGAN model, potentially reflecting its data-driven stochastic formulation. The CGAN model demonstrates a comparatively lower total aleatory uncertainty, suggesting an improved characterization of variability compared to the Kotha2022 model.

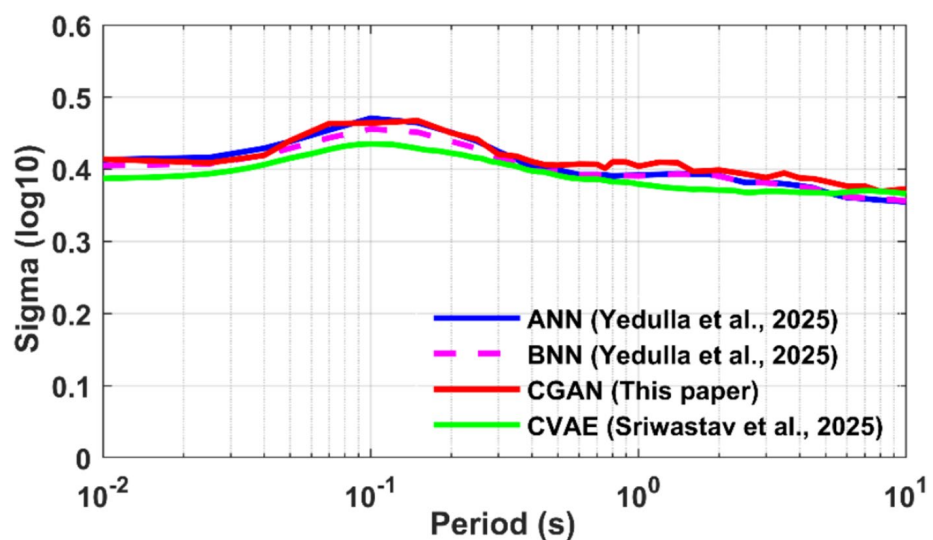
Further, Fig. 19 presents a comparative analysis of aleatory uncertainty across four models—ANN and BNN (Yedulla et al. 2025), CVAE (Sriwastav et al. 2025), and CGAN (this study)—all developed using the same dataset. All the models lead to nearly the same estimate, with values ranging from 0.35 to 0.45 across

periods. CGAN performs comparably to BNN and ANN at all periods. CVAE consistently achieves comparatively lower sigma values, likely due to its latent-space regularization. Overall, CGAN balances realism and uncertainty modeling effectively, offering competitive performance while preserving flexibility in simulating diverse ground motion spectra. Discussion on epistemic uncertainty is presented in the following section.

6.3.2 Epistemic uncertainty

A key advantage of CGAN is its ability to generate an ensemble of ground motions instead of a single point estimate. Figure 20 presents ensemble predictions of horizontal spectral acceleration across four diverse seismic scenarios, highlighting the CGAN model's ability to capture epistemic uncertainties. Each panel reflects a different combination of magnitude, distance, and site condition, resulting in varied levels of spectral amplitude and dispersion. Overall, the predicted ensembles exhibit realistic variability, with the 95% confidence intervals encompassing the recorded ground motions in all cases. The median predictions closely follow the observed trends, indicating the model's robustness across a wide range of input conditions. The spread of predictions is generally wider for larger events and stiffer sites, where greater variability is expected, and narrower for smaller magnitudes and softer soils due to the relative number of records available.

Fig. 19 Aleatory uncertainty of the response spectra for the ESM data computed using ANN, BNN, CGAN and CVAE



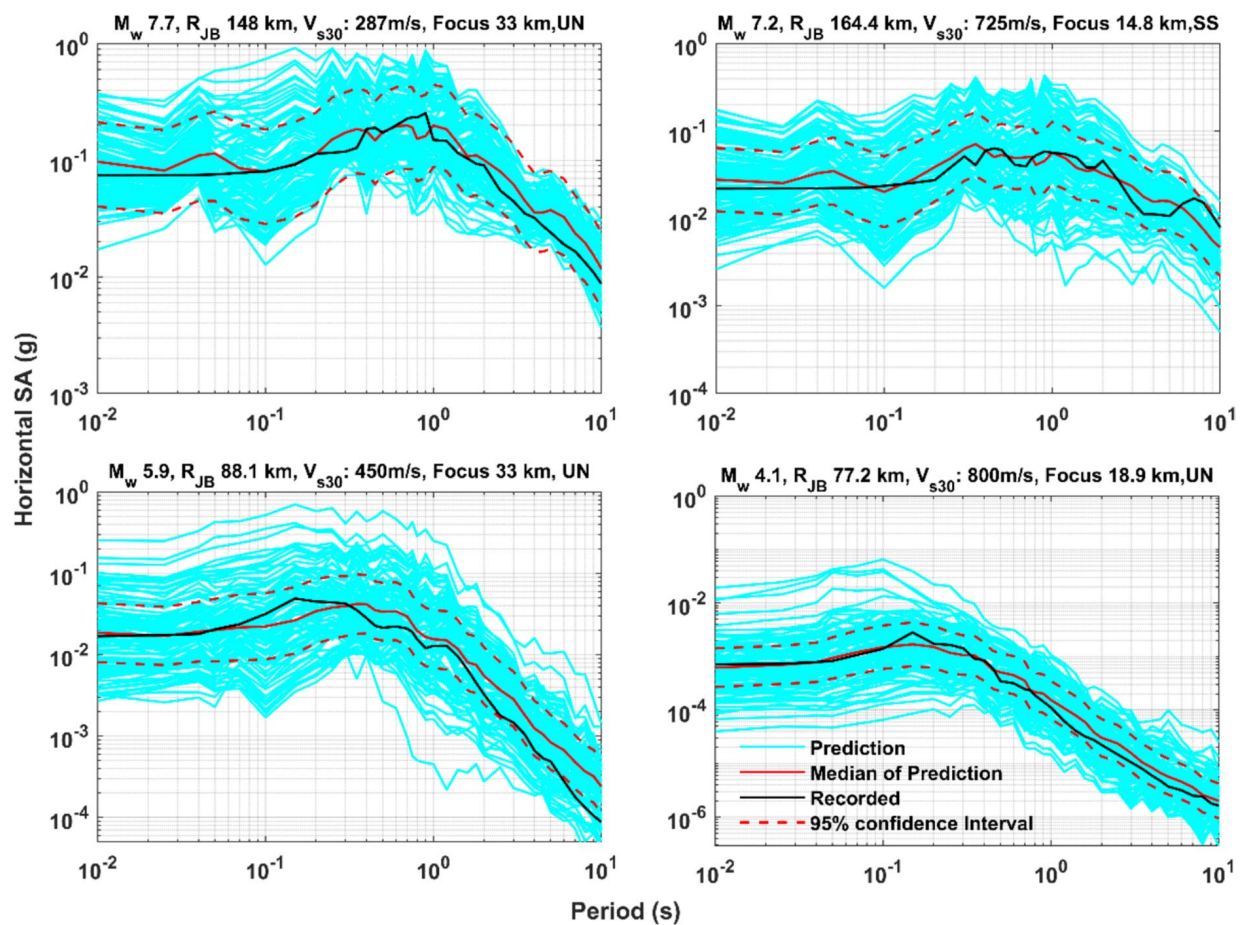


Fig. 20 100 ensemble outputs for 4 sample ground motions using the proposed CGAN model, the median spectra of the ensemble prediction, the spectra of the recorded ground motions and the 95% confidence interval bounds

Considering “M” samples are generated for individual (n^{th}) ground motions using CGAN, the epistemic standard deviation of any n^{th} ground motion is denoted by $\sigma_{n-\text{epistemic}}$. The variation of $\sigma_{n-\text{epistemic}}$ with the seismological parameters across four spectral periods ($T=0.01$ to 10.0 s), presented in Figure 27 in the Appendix. $\sigma_{n-\text{epistemic}}$ is stable for M_w 4.5–6.5 but rises at $M_w > 6.5$, especially at long periods, due to limited data for large events. Higher $\sigma_{n-\text{epistemic}}$ at <20 km and >250 km is the result of sparse recordings and path effects. Higher $\sigma_{n-\text{epistemic}}$, particularly for $V_{s30} > 1300$ m/s is seen due to underrepresented stiff sites. Further, higher uncertainty for focal depths >30 km is linked to shallow-crustal dominance of records. Little dependence on fault mechanism is seen. Further, the contour plots of epistemic uncertainty mapped over M_w

$-R_{JB}$ grids for the same four periods are presented in Figure 28 (in the Appendix). These contour highlights where the model exhibits higher variance in predictions, which correlates strongly with areas of limited training data. Similar contours, however, classified for different site conditions, are presented between Figures 29, 30 and 31. Uncertainty is generally moderate for $M_w < 6.5$ and $R_{JB} < 150$ km, reflecting dense data coverage. It increases for lower magnitudes at far distances and larger magnitudes, owing to data scarcity and event-to-event variability. Period dependence reveals low uncertainty at short periods (0.01 s), rising at 0.1–1.0 s. For 10.0 s, the spectral ordinates are often comparatively small, leading to lower uncertainty. However, it increases for larger M_w due to data scarcity. The trend is nearly the same for all site conditions. Overall, this

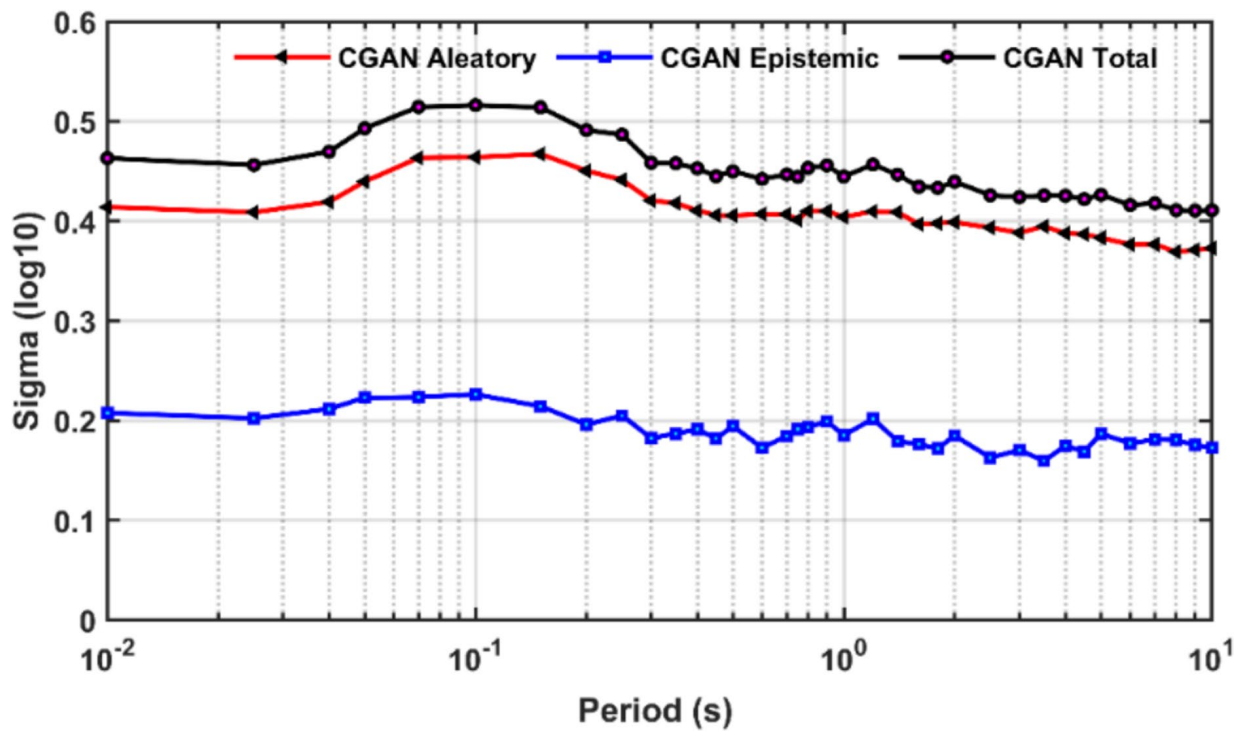


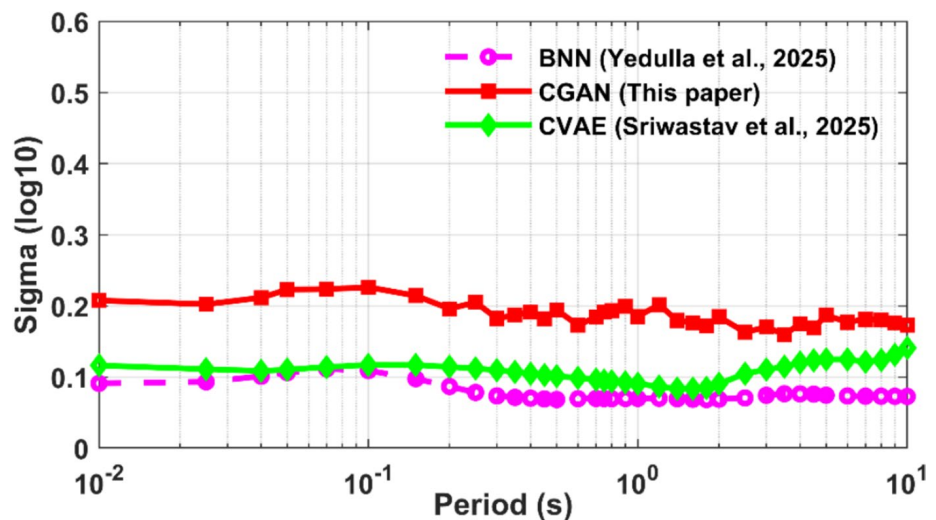
Fig. 21 Aleatory uncertainty, epistemic uncertainty and the total uncertainty computed using CGAN

adversarial noise-induced variability in the CGAN serves as a proxy for epistemic uncertainty, remaining low in data-rich regimes but increasing in sparse regions, as demonstrated in the figures.

For simplicity, the overall epistemic uncertainty ($\sigma_{n\text{-epistemic}}$) is computed as the mean of all the standard deviations for all the N predictions as follows.

$$\sigma_{n\text{-epistemic}}^2(T) = \frac{1}{N} \sum_{n=1}^N \sigma_{n\text{-epistemic}}^2(T); \sigma_{n\text{-epistemic}}^2(T) = \frac{1}{M-1} \sum_{m=1}^M (y_{m,n} - \bar{y}_n)^2 \quad (10)$$

Fig. 22 Epistemic uncertainty for the ESM data computed using BNN, CGAN and CVAE



Here N is 11675 and M is taken as 100 despite that a 400-ensemble prediction is recommended. This is because, beyond this threshold, increases in simulation count yielded only small changes in the estimated standard deviation. The comparison of the standard deviations, particularly aleatory, epistemic, and total standard deviations, is presented in Fig. 21, where total epistemic uncertainty is computed using the SRSS of the aleatory and epistemic uncertainty [Eq. (11)].

$$\sigma_{total} = \sqrt{\sigma_{epistemic}^2 + \sigma_{aleatory}^2} \quad (11)$$

A comparative analysis of epistemic uncertainty using BNN (Yedulla et al. 2025), CVAE (Sriwastav et al. 2025), and CGAN (this study), all developed using the same dataset, is presented in Fig. 22. The results highlight variations in epistemic uncertainty across models: BNNs yield the narrowest uncertainty bands, CVAEs produce moderately higher estimates, while CGANs exhibit the broadest. These differences may be attributed to the source of epistemic uncertainty. While BNN extracts the uncertainty from model weights and CVAE attains uncertainty by learning a latent variable model, the source of uncertainty in CGAN is the random noise in the generator. The CGAN model exhibits higher epistemic uncertainty compared to the other models. Consequently, the total standard deviation for the CGAN model is higher. This increased total standard deviation underscores the CGAN model's greater sensitivity to the inherent variability introduced by its random noise generator. Incorporating epistemic uncertainty into the GMM using CGAN could provide a viable

alternative to the traditional logic-tree approach for site-specific seismic hazard assessment. However, this topic warrants a separate study and is not discussed further in this work.

7 Evaluating median CGAN predictions against state-of-the-art GMMs

This section evaluates the proposed CGAN model against state-of-the-art GMMs in the existing literature. Among the several GMMs developed using ESM data or its subsets, the most recent and widely used models for the horizontal spectral ordinate include Bindi et al. (2014), Akkar et al. (2014), Laouami et al. (2018), Kotha et al. (2020, 2022), and Sreenath et al. (2023). The comparison of the CGAN prediction is conducted with these models, as well as other GMMs developed using global databases (Campbell and Bozorgnia, 2014, and Boore et al. 2014).

Several GMMs are developed for different sub-regions in Europe. Among these models, Bindi et al. (2014) used a large database covering Europe and the Middle East, utilizing the RESORCE strong motion database to develop a GMM for 5% damped response spectra. The database, however, mostly dominates Turkey and Italy ground motions. Similarly, Akkar et al. (2014) utilized recordings from shallow active crustal regions in Europe and the Middle East. Laouami et al. (2018) developed the GMM using strong motions mainly from Algeria, Europe, and the USA. Kotha et al. (2020) presented a GMM for the horizontal component utilizing a comparatively large dataset from the

Table 4 Features of the GMMs selected for the comparison with the proposed CGAN model for predicting horizontal spectral acceleration (S_a)

GMM region	M_w range	Distance range (km)	Style of faulting	Site class, V_{s30}	Period range (sec)
Bindi et al. 2014	$4.0 \leq M_w \leq 7.6$	0–300 km	NF, TF, SS, UN	EC8 site classes	0.02–3.00
Akkar et al. 2014	$4.0 \leq M_w \leq 8.0$	0–200 km	NF, SS, TF	180–1200 m/s	0.01–4.00
Laouami et al. 2018	$3.0 \leq M_w \leq 7.4$	5 to 150 km	TF	Soil class based on Zhao et al. (2006)	0.02–4.00
Kotha et al. 2022	$3.1 \leq M_w \leq 7.4$	0–545 km	NF, TF, SS, UN	90–3000 m/s	0.001–4.00
Sreenath et al. 2023	$3.2 \leq M_w \leq 7.4$	0–545 km	–	90–2930 m/s	0.01–8.00
CB2014	$3.0 \leq M_w \leq 7.9 \leq 7.98.5$	0–500 km	NF, TF, SS	150–1500 m/s	0.01–10
Boore et al. 2014	$3.0 \leq M_w \leq 7.9$	0–400 km	NF, TF, SS	150–2016 m/s	0.01–10
This study	$4.0 \leq M_w \leq 8.0$	0–400 km	NF, TF, SS, UN	180–2000 m/s	0.01–10

**NF Normal Fault, TF Thrust Fault, SS Strike Slip, UN Unknown

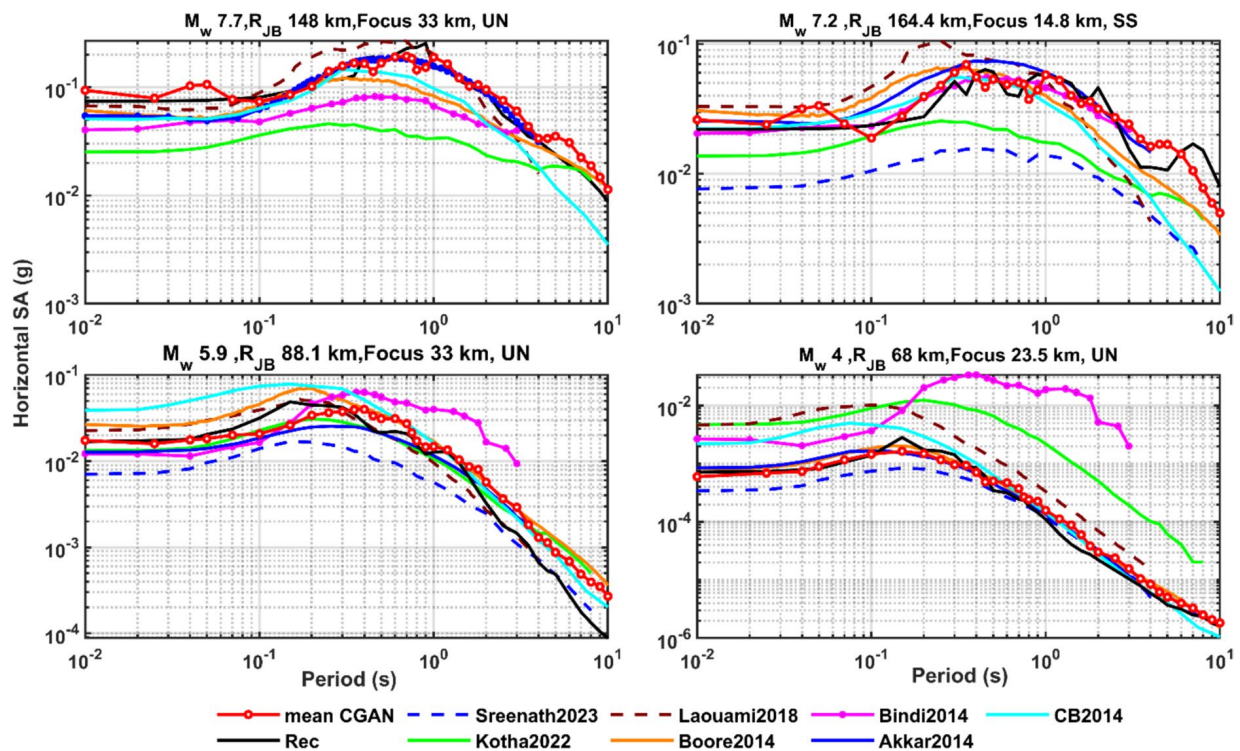


Fig. 23 Comparison of the GMM predicted for horizontal component in the current study with the past models developed for the European region

ESM database. This model utilized additional random effect groups other than inter-event, inter-site, such as attenuation regions and tectonic localities. Kotha et al. (2022) updated Kotha et al. (2020) by including near-source strong motion (NESS) database to rectify magnitude scaling and revised the fixed-effect regression coefficients. Sreenath et al. (2023) utilized the same dataset as wells as random effects and developed the hybrid non-parametric model using various machine learning techniques. Campbell and Bozorgnia (2014) (CB2014) developed a GMM for the horizontal component with average horizontal components as the intensity measure using PEER NGA-West2 database. This model is most comprehensive yet complex and is used throughout the world for seismic hazard estimation. Boore et al. (2014), in similar lines, have been widely used for PSHA. Salient details of these GMMs are presented in Table 4.

Figure 23 compares the median CGAN prediction of this paper (median of 100 predictions) with that of GMM such as Bindi et al. (2014), Campbell and Bozorgnia (2014), Boore et al. (2014), Laouami et al. (2018), Kotha et al. (2022), Sreenath et al. (2023).

Sample comparisons are provided across various scenarios defined by M_w , R_{JB} and V_{s30} , focal depth, and focal mechanisms. Also included in the comparison is the response spectra of the recorded ground motions (denoted by ‘rec’) for the same scenario as a benchmark. Despite the fact that the median predictions cannot be directly compared with the spectra of individual recorded ground motions, these comparisons provide a qualitative evaluation of the proposed model. The median prediction by Bindi et al. (2014) for sample scenario such as M_w 7.2 is close to its recorded counterpart. However, the predictions for almost all other scenarios deviate, indicating a localized and biased dataset. The median predictions from Akkar et al. (2014), CB2014, and the CGAN model are close to the spectral ordinates of the recorded ground motions for most scenarios and periods, with the exception of CB2014, where the median predictions are comparatively lower at higher periods. The prediction of Boore et al. (2014) closely align with the recorded spectra and the CGAN model predictions for lower magnitude earthquakes. However, it is observed to deviate for higher magnitudes.

The predictions by Laouami et al. (2018) are relatively higher compared to most of the GMMs used for comparison, indicating a high positive bias in the model. In contrast, the predictions by Sreenath et al. (2023) are lower for all scenarios when compared to other GMMs as well as the recorded ground motions. The prediction by Sreenath et al. (2023) is not included in the first panel of Fig. 23, as its applicable magnitude range is limited to M_w 7.4. The predictions by Kotha et al. (2022) are also on the lower side, particularly for higher magnitude earthquakes, though their predictions for intermediate magnitudes are close to those of other GMMs. The similarity of predictions for all GMMs at intermediate magnitudes is likely due to the large number of recordings available in this range. The predictions of Kotha et al. (2022) are observed to be quite on the higher side for lower magnitude earthquakes. The proposed model, on the other hand, demonstrated better alignment with the recorded ground motion for all the scenarios at all periods. It may also be inferred from the overall comparison that the non-parametric CGAN model can efficiently match spectra of the recorded ground motions as well as predict at par with complex and exhaustive GMMs such as Campbell and Bozorgnia (2014) and Boore et al. (2014).

8 Summary and conclusion

This paper explores the potential of deep learning technique, conditional generative adversarial network (CGAN), in the development of ground motion models (GMMs). Compared to deterministic models, CGAN carries an advantage of generating ensemble predictions rather than point estimates. The CGAN model is trained on spectral accelerations at 37 periods for horizontal components between 0–10 s using 11,675 sets of records from the Engineering Strong Motion (ESM2.0) database. The model is applicable in the following range: M_w 4.0–7.8, $R_{JB} < 400$ km, focal depth (d) < 40 km and V_{s30} between 180–1300 m/s.

The CGAN model demonstrated competitive performance conventional GMMs in terms of median prediction. Further, the ability of CGAN to simulate random realizations offers a clear distinction from conventional machine learning techniques. Mixed-effects regression was conducted to quantify standard deviations associated with inter-event, inter-site, inter-locality, and inter-regional variability. When compared to the regression-based model of Kotha et al. (2022), the CGAN showed

a lower total aleatory uncertainty. Comparison with an Artificial Neural Network ANN, Bayesian Neural Network (BNN) model (Yedulla et al. 2025) and Conditional Variational Autoencoder (CVAE) model (Sriwastav et al. 2025) indicated comparable aleatory variability; however, the CGAN exhibited higher epistemic uncertainty than both the BNN and CVAE models. This difference is attributed to the different sources of epistemic uncertainty: CGAN's stems from input noise (analogous to source uncertainty), BNN's arises from posterior model weights (analogous to path uncertainty) whereas CVAE's arises from latent variable distribution (analogous to site uncertainty). Despite their differing interpretations, CGANs provide a valuable framework for modeling epistemic uncertainty in ground motion simulations. Overall, CGANs show strong potential for seismological applications, particularly in representing uncertainty in GMM development. This study also suggests that CGANs may serve as a viable alternative to logic-tree frameworks for incorporating epistemic uncertainty in site-specific seismic hazard assessments. Furthermore, the CGAN's success in spectral prediction encourages future work on generating synthetic time histories.

Author contributions RK: Methodology design, Literature review, data curation, validation, visualization, and original draft preparation. JY: Investigation, result interpretation, preparation of figures and tables, and editing of the manuscript. PB: Assistance in software development, data preprocessing, formal analysis. STG: Conceptualization, Supervision, project administration, and critical revision of the manuscript for intellectual content. All authors have read and approved the final version of the manuscript.

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Data availability ESM2.0 flatfile is available at https://esm-db.eu/#/products/flat_file, and the database is available at <https://esm.mi.ingv.it/flatfile-2018>. No additional datasets were generated or analysed during the current study.

Code availability The Python code for generating 5% damped acceleration response spectra for using the CGAN model is provided in <https://figshare.com/s/284c90f5f9868473bb3b>. The Python code for CGAN used in this study is available upon request.

Declarations

Competing interests The authors declare no competing interests.